APPENDIX B

Instruction Set Summary

The instruction set summary contains a complete alphabetical listing of the entire 8086–Pentium 4 instruction set. The coprocessor and MMX instructions are listed in Chapter 14 and are not repeated in this appendix. The SIMD instructions appear at the end of this appendix after the main instruction set summary.

Each instruction entry lists the mnemonic opcode plus a brief description of the purpose of the instruction. Also listed is the binary machine language coding of each instruction and any other data required to form the instruction, such as the displacement or immediate data. Listed to the right of each binary machine language version of the instruction are the flag bits and any change that might occur for the instruction. The flags are described in the following manner: A blank indicates no effect or change; a ? indicates a change with an unpredictable outcome; a * indicates a change with a predictable outcome; a 1 indicates the flag is set; and a 0 indicates that the flag is cleared. If the flag bits ODITSZAPC are not illustrated with an instruction, the instruction does not modify any of these flags.

Before the instruction listing begins, some information about the bit settings in binary machine language versions of the instructions is presented. Table B–1 lists the modifier bits, coded as OO in the instruction listing.

Table B–2 lists the memory-addressing modes available using a register field coding of mmm. This table applies to all versions of the microprocessor, as long as the operating mode is 16 bits.

Table B–3 lists the register selections provided by the rrr field in an instruction. This table includes the register selections for 8-, 16-, and 32-bit registers.

Table B–4 lists the segment register bit assignment (rrr) found with the MOV, PUSH, and POP instructions.

TABLE B–1 The modifier bits, coded as oo in the instruction listing.

oo	*Function*
00	If mmm = 110, a displacement follows the opcode; otherwise no displacement is used.
01	An 8-bit signed displacement flows the opcode.
10	A 16- or 32-bit signed displacement follows the opcode.
11	mmm specifies a register instead of an addressing mode.

TABLE B–2 The 16-bit register/memory (mmm) field description.

mmm	*16-Bit*
000	DS:[BX+SI]
001	DS:[BX+DI]
010	SS:[BP+SI]
011	SS:[BP+DI]
100	DS:[SI]
101	DS:[DI]
110	SS:[BP]
111	DS:[BX]

TABLE B–3 The register (rrr) field.

rrr	*W=0*	*W=1 (16-Bit)*	*W=1 (32-Bit)*
000	AL	AX	EAX
001	CL	CX	ECX
010	DL	DX	EDX
011	BL	BX	EBX
100	AH	SP	ESP
101	CH	BP	EBP
110	DH	SI	ESI
111	BH	DI	EDI

TABLE B–4 Register field assignments (rrr) for the segment registers.

rrr	*Segment Register*
000	ES
001	CS
010	SS
011	DS
100	FS
101	GS

When the 80386–Core2 are used, some of the definitions provided in Table B–1 through B–3 change. See Tables B–5 and B–6 for these changes as they apply to the 80386–Core2 microprocessors.

TABLE B–5 Index registers specified by rrr when the 80386–Core2 are operated in 32-bit mode.

rrr	*Index Register*
000	DS:[EAX]
001	DS:[ECX]
010	DS:[EDX]
011	DS:[EBX]
100	(see Table B–6)
101	SS:[EBP]
110	DS:[ESI]
111	DS:[EDI]

TABLE B–6 Possible combinations of oo, mmm, and rrr for the 80386–Core2 microprocessors using 32-bit addressing.

oo	*mmm*	*rrr (Base in Scaled-Index Byte)*	*Addressing Mode*
00	000	—	DS:[EAX]
00	001	—	DS:[ECX]
00	010	—	DS:[EDX]
00	011	—	DS:[EBX]
00	100	000	DS:[EAX+scaled index]
00	100	001	DS:[ECX+scaled index]
00	100	010	DS:[EDX+scaled index]
00	100	011	DS:[EBX+scaled index]
00	100	100	SS:[ESP+scaled index]
00	100	101	DS:[disp32+scaled index]
00	100	110	DS:[ESI+scaled index]
00	100	111	DS:[EDI+scaled index]
00	101	—	DS:disp32
00	110	—	DS:[ESI]
00	111	—	DS:[EDI]
01	000	—	DS:[EAX+disp8]
01	001	—	DS:[ECX+disp8]
01	010	—	DS:[EDX+disp8]
01	011	—	DS:[EBX+disp8]
01	100	000	DS:[EAX+scaled index+disp8]
01	100	001	DS:[ECX+scaled index+disp8]
01	100	010	DS:[EDX+scaled index+disp8]
01	100	011	DS:[EBX+scaled index+disp8]
01	100	100	SS:[ESP+scaled index+disp8]
01	100	101	SS:[EBP+scaled index+disp8]
01	100	110	DS:[ESI+scaled index+disp8]
01	100	111	DS:[EDI+scaled index+disp8]
01	101	—	SS:[EBP+disp8]
01	110	—	DS:[ESI+disp8]
01	111	—	DS:[EDI+disp8]
10	000	—	DS:[EAX+disp32]
10	001	—	DS:[ECX+disp32]
10	010	—	DS:[EDX+disp32]
10	011	—	DS:[EBX+disp32]
10	100	000	DS:[EAX+scaled index+disp32]
10	100	001	DS:[ECX+scaled index+disp32]
10	100	010	DS:[EDX+scaled index+disp32]
10	100	011	DS:[EBX+scaled index+disp32]
10	100	100	SS:[ESP+scaled index+disp32]
10	100	101	SS:[EBP+scaled index+disp32]
10	100	110	DS:[ESI+scaled index+disp32]
10	100	111	DS:[EDI+scaled index+disp32]
10	101	—	SS:[EBP+disp32]
10	110	—	DS:[ESI+disp32]
10	111	—	DS:[EDI+disp32]

Note: disp8 = 8-bit displacement and disp32 = 32-bit displacement.

In order to use the scaled-index addressing modes listed in Table B–6, code oo and mmm in the second byte of the opcode. The scaled-index byte is usually the third byte and contains three fields. The leftmost two bits determine the scaling factor (00 = ×1, 01 = ×2, 10 = ×4, or 11 = ×8). The next three bits toward the right contain the scaled-index register number (this is obtained from Table B–5). The rightmost three bits are from the mmm field listed in Table B–6. For example, the MOV AL,[EBX+2*ECX] instruction has a scaled-index byte of 01 001 011, where 01 = X_2, 001 = ECX, and 011 = EBX.

Some instructions are prefixed to change the default segment or to override the instruction mode. Table B–7 lists the segment and instruction mode override prefixes with append at the beginning of an instruction if they are used to form the instruction. For example, the MOV AL,ES:[BX] instruction used the extra segment because of the override prefix ES:.

In the 8086 and 8088 microprocessors, the effective address calculation required additional clocks that are added to the times in the instruction set summary. These additional times are listed in Table B–8. No such times are added to the 80286–Core2. Note that the instruction set summary does not include clock times for the Pentium Pro through the Core2. Intel has not released these times and has decided that the RDTSC instruction can be used to have the microprocessor count the number of clocks required for a given application. Even though the timings do not appear for these new microprocessors, they are very similar to the Pentium, which can be used as a guide.

TABLE B–7 Override prefixes.

Prefix Byte	*Purpose*
26H	ES: segment override
2EH	CS: segment override
36H	SS: segment override
3EH	DS: segment override
64H	FS: segment override
65H	GS: segment override
66H	Memory operand instruction mode override
67H	Register operand instruction mode override

TABLE B–8 Effective address calculations for the 8086 and 8088 microprocessors.

Type	*Clocks*	*Example Instruction*
Base or index	5	MOV CL,[DI]
Displacement	3	MOV AL,DATA1
Base plus index	7	MOV AL,[BP+SI]
Displacement plus base or index	9	MOV DH,[DI+20H]
Base plus index plus displacement	11	MOV CL,[BX+DI+2]
Segment override	ea + 2	MOV AL,ED:[DI]

INSTRUCTION SET SUMMARY

AAA ASCII adjust AL after addition

00110111 — O D I T S Z A P C: ? ? ? ? * ? * ?

Example	Microprocessor	Clocks
AAA	8086	8
	8088	8
	80286	3
	80386	4
	80486	3
	Pentium–Core2	

AAD ASCII adjust AX before division

11010101 00001010 — O D I T S Z A P C: ? ? * * ? * ?

Example	Microprocessor	Clocks
AAD	8086	60
	8088	60
	80286	14
	80386	19
	80486	14
	Pentium–Core2	10

AAM ASCII adjust AX after multiplication

11010100 00001010 — O D I T S Z A P C: ? ? * * ? * ?

Example	Microprocessor	Clocks
AAM	8086	83
	8088	83
	80286	16
	80386	17
	80486	15
	Pentium–Core2	18

AAS ASCII adjust AL after subtraction

00111111

O D I T S Z A P C
? ? ? * ? *

Example	Microprocessor	Clocks
AAS	8086	8
	8088	8
	80286	3
	80386	4
	80486	3
	Pentium–Core2	3

ADC Addition with carry

000100dw oorrrmmm disp

O D I T S Z A P C
* * * * *

Format	Examples	Microprocessor	Clocks
ADC reg,reg	ADC AX,BX ADC AL,BL ADC EAX,EBX ADC CX,SI ADC ESI,EDI	8086	3
		8088	3
		80286	3
		80386	3
		80486	1
		Pentium–Core2	1 or 3
ADC mem,reg	ADC DATAY,AL ADC LIST,SI ADC DATA2[DI],CL ADC [EAX],BL ADC [EBX+2*ECX],EDX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3

ADC reg,mem	ADC BL,DATA1 ADC SI,LIST1 ADC CL,DATA2[SI] ADC CX,[ESI] ADC ESI,[2*ECX]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	6
		80486	2
		Pentium–Core2	1 or 2
100000sw oo010mmm disp data Format	Examples	Microprocessor	Clocks
ADC reg,imm	ADC CX,3 ADC DI,1AH ADC DL,34H ADC EAX,12345 ADC CX,1234H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3
ADC mem,imm	ADC DATA4,33 ADC LIST,'A' ADC DATA3[DI],2 ADC BYTE PTR[EBX],3 ADC WORD PTR[DI],669H	8086	17 + ea
		8088	23 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
ADC acc,imm	ADC AX,3 ADC AL,1AH ADC AH,34H ADC EAX,2 ADC AL,'Z'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

ADD Addition

000000dw oorrrmmm disp

O D I T S Z A P C
* * * * * *

Format	Examples	Microprocessor	Clocks
ADD reg,reg	ADD AX,BX ADD AL,BL ADD EAX,EBX ADD CX,SI ADD ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3
ADD mem,reg	ADD DATAY,AL ADD LIST,SI ADD DATA6[DI],CL ADD [EAX],CL ADD [EDX+4*ECX],EBX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
ADD reg,mem	ADD BL,DATA2 ADD SI,LIST3 ADD CL,DATA2[DI] ADD CX,[EDI] ADD ESI,[ECX+200H]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	6
		80486	2
		Pentium–Core2	1 or 2

100000sw oo000mmm disp data

Format	Examples	Microprocessor	Clocks
ADD reg,imm	ADD CX,3 ADD DI,1AH ADD DL,34H ADD EDX,1345H ADD CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3

ADD mem,imm	ADD DATA4,33 ADD LIST,'A' ADD DATA3[DI],2 ADD BYTE PTR[EBX],3 ADD WORD PTR[DI],669H	8086	17 + ea
		8088	23 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
ADD acc,imm	ADD AX,3 ADD AL,1AH ADD AH,34H ADD EAX,2 ADD AL,'Z'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

AND Logical AND

001000dw oorrrmmm disp

O	D	I	T	S	Z	A	P	C
0				*	*	?	*	0

Format	Examples	Microprocessor	Clocks
AND reg,reg	AND CX,BX AND DL,BL AND ECX,EBX AND BP,SI AND EDX,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3
AND mem,reg	AND BIT,AL AND LIST,DI AND DATAZ[BX],CL AND [EAX],BL AND [ESI+4*ECX],EDX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3

AND reg,mem	AND BL,DATAW AND SI,LIST AND CL,DATAQ[SI] AND CX,[EAX] AND ESI,[ECX+43H]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	6
		80486	2
		Pentium–Core2	1 or 2
100000sw oo100mmm disp data Format	Examples	Microprocessor	Clocks
AND reg,imm	AND BP,1 AND DI,10H AND DL,34H AND EBP,1345H AND SP,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3
AND mem,imm	AND DATA4,33 AND LIST,'A' AND DATA3[DI],2 AND BYTE PTR[EBX],3 AND DWORD PTR[DI],66H	8086	17 + ea
		8088	23 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
AND acc,imm	AND AX,3 AND AL,1AH AND AH,34H AND EAX,2 AND AL,'r'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

ARPL Adjust requested privilege level

01100011 oorrrmmm disp

O D I T S Z A P C
*

Format	Examples	Microprocessor	Clocks
ARPL reg,reg	ARPL AX,BX ARPL BX,SI ARPL AX,DX ARPL BX,AX ARPL SI,DI	8086	—
		8088	—
		80286	10
		80386	20
		80486	9
		Pentium–Core2	7
ARPL mem,reg	ARPL DATAY,AX ARPL LIST,DI ARPL DATA3[DI],CX ARPL [EBX],AX ARPL [EDX+4*ECX],BP	8086	—
		8088	—
		80286	11
		80386	21
		80486	9
		Pentium–Core2	7

BOUND Check array against boundary

01100010 oorrrmmm disp

Format	Examples	Microprocessor	Clocks
BOUND reg,mem	BOUND AX,BETS BOUND BP,LISTG BOUND CX,DATAX BOUND BX,[DI] BOUND SI,[BX+2]	8086	—
		8088	—
		80286	13
		80386	10
		80486	7
		Pentium–Core2	8

BSF Bit scan forward

00001111 10111100 oorrrmmm disp

O D I T S Z A P C
? ? * ? ? ?

Format	Examples	Microprocessor	Clocks
BSF reg,reg	BSF AX,BX BSF BX,SI BSF EAX,EDX BSF EBX,EAX BSF SI,DI	8086	—
		8088	—
		80286	—
		80386	10 + 3n
		80486	6–42
		Pentium–Core2	6–42
BSF reg,mem	BSF AX,DATAY BSF SI,LIST BSF CX,DATA3[DI] BSF EAX,[EBX] BSF EBP,[EDX+4*ECX]	8086	—
		8088	—
		80286	—
		80386	10 + 3n
		80486	7–43
		Pentium–Core2	6–43

BSR Bit scan reverse

00001111 10111101 oorrrmmm disp

O D I T S Z A P C
? ? * ? ? ?

Format	Examples	Microprocessor	Clocks
BSR reg,reg	BSR AX,BX BSR BX,SI BSR EAX,EDX BSR EBX,EAX BSR SI,DI	8086	—
		8088	—
		80286	—
		80386	10 + 3n
		80486	6–103
		Pentium–Core2	7–71

BSR reg,mem	BSR AX,DATAY BSR SI,LIST BSR CX,DATA3[DI] BSR EAX,[EBX] BSR EBP,[EDX+4*ECX]	8086	—
		8088	—
		80286	—
		80386	10 + 3n
		80486	7–104
		Pentium–Core2	7–72

BSWAP Byte swap

00001111 11001rrr

Format	Examples	Microprocessor	Clocks
BSWAP reg32	BSWAP EAX BSWAP EBX BSWAP EDX BSWAP ECX BSWAP ESI	8086	—
		8088	—
		80286	—
		80386	—
		80486	1
		Pentium–Core2	1

BT Bit test

00001111 10111010 oo100mmm disp data

O D I T S Z A P C
*

Format	Examples	Microprocessor	Clocks
BT reg,imm8	BT AX,2 BT CX,4 BT BP,10H BT CX,8 BT BX,2	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	4

BT mem,imm8	BT DATA1,2 BT LIST,2 BT DATA2[DI],3 BT [EAX],1 BT FROG,6	8086	—
		8088	—
		80286	—
		80386	6
		80486	3
		Pentium–Core2	4
00001111 10100011 disp Format	Examples	Microprocessor	Clocks
BT reg,reg	BT AX,CX BT CX,DX BT BP,AX BT SI,CX BT EAX,EBX	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	4 or 9
BT mem,reg	BT DATA4,AX BT LIST,BX BT DATA3[DI],CX BT [EBX],DX BT [DI],DI	8086	—
		8088	—
		80286	—
		80386	12
		80486	8
		Pentium–Core2	4 or 9

BTC Bit test and complement

00001111 10111010 oo111mmm disp data — O D I T S Z A P C *

Format	Examples	Microprocessor	Clocks
BTC reg,imm8	BTC AX,2 BTC CX,4 BTC BP,10H BTC CX,8 BTC BX,2	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 8
BTC mem,imm8	BTC DATA1,2 BTC LIST,2 BTC DATA2[DI],3 BTC [EAX],1 BTC FROG,6	8086	—
		8088	—
		80286	—
		80386	7 or 8
		80486	8
		Pentium–Core2	8

00001111 10111011 disp

Format	Examples	Microprocessor	Clocks
BTC reg,reg	BTC AX,CX BTC CX,DX BTC BP,AX BTC SI,CX BTC EAX,EBX	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 13
BTC mem,reg	BTC DATA4,AX BTC LIST,BX BTC DATA3[DI],CX BTC [EBX],DX BTC [DI],DI	8086	—
		8088	—
		80286	—
		80386	13
		80486	13
		Pentium–Core2	7 or 13

BTR Bit test and reset			
00001111 10111010 oo110mmm disp data		O D I T S Z A P C *	
Format	Examples	Microprocessor	Clocks
BTR reg,imm8	BTR AX,2 BTR CX,4 BTR BP,10H BTR CX,8 BTR BX,2	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 8
BTR mem,imm8	BTR DATA1,2 BTR LIST,2 BTR DATA2[DI],3 BTR [EAX],1 BTR FROG,6	8086	—
		8088	—
		80286	—
		80386	8
		80486	8
		Pentium–Core2	7 or 8
00001111 10110011 disp Format	Examples	Microprocessor	Clocks
BTR reg,reg	BTR AX,CX BTR CX,DX BTR BP,AX BTR SI,CX BTR EAX,EBX	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 13
BTR mem,reg	BTR DATA4,AX BTR LIST,BX BTR DATA3[DI],CX BTR [EBX],DX BTR [DI],DI BTC [DI],DI	8086	—
		8088	—
		80286	—
		80386	13
		80486	13
		Pentium–Core2	7 or 13

BTS Bit test and set

00001111 10111010 oo101mmm disp data — O D I T S Z A P C *

Format	Examples	Microprocessor	Clocks
BTS reg,imm8	BTS AX,2 BTS CX,4 BTS BP,10H BTS CX,8 BTS BX,2	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 8
BTS mem,imm8	BTS DATA1,2 BTS LIST,2 BTS DATA2[DI],3 BTS [EAX],1 BTS FROG,6	8086	—
		8088	—
		80286	—
		80386	8
		80486	8
		Pentium–Core2	7 or 8

00001111 10101011 disp

Format	Examples	Microprocessor	Clocks
BTS reg,reg	BTS AX,CX BTS CX,DX BTS BP,AX BTS SI,CX BTS EAX,EBX	8086	—
		8088	—
		80286	—
		80386	6
		80486	6
		Pentium–Core2	7 or 13
BTS mem,reg	BTS DATA4,AX BTS LIST,BX BTS DATA3[DI],CX BTS [EBX],DX BTS [DI],DI	8086	—
		8088	—
		80286	—
		80386	13
		80486	13
		Pentium–Core2	7 or 13

CALL	Call procedure (subroutine)		
11101000 disp Format	Examples	Microprocessor	Clocks
CALL label (near)	CALL FOR_FUN CALL HOME CALL ET CALL WAITING CALL SOMEONE	8086	19
		8088	23
		80286	7
		80386	3
		80486	3
		Pentium–Core2	1
10011010 disp Format	Examples	Microprocessor	Clocks
CALL label (far)	CALL FAR PTR DATES CALL WHAT CALL WHERE CALL FARCE CALL WHOM	8086	28
		8088	36
		80286	13
		80386	17
		80486	18
		Pentium–Core2	4
11111111 oo010mmm Format	Examples	Microprocessor	Clocks
CALL reg (near)	CALL AX CALL BX CALL CX CALL DI CALL SI	8086	16
		8088	20
		80286	7
		80386	7
		80486	5
		Pentium–Core2	2

CALL mem (near)	CALL ADDRESS CALL NEAR PTR [DI] CALL DATA1 CALL FROG CALL ME_NOW	8086	21 + ea
		8088	29 + ea
		80286	11
		80386	10
		80486	5
		Pentium–Core2	2
11111111 oo011mmm Format	Examples	Microprocessor	Clocks
CALL mem (far)	CALL FAR_LIST[SI] CALL FROM_HERE CALL TO_THERE CALL SIXX CALL OCT	8086	16
		8088	20
		80286	7
		80386	7
		80486	5
		Pentium–Core2	2

CBW Convert byte to word (AL ⇒ AX)

10011000 Example	Microprocessor	Clocks
CBW	8086	2
	8088	2
	80286	2
	80386	3
	80486	3
	Pentium–Core2	3

CDQ Convert doubleword to quadword (EAX ⇒ EDX:EAX)

11010100 00001010

Example	Microprocessor	Clocks
CDQ	8086	—
	8088	—
	80286	—
	80386	2
	80486	2
	Pentium–Core2	2

CLC Clear carry flag

11111000 O D I T S Z A P C
0

Example	Microprocessor	Clocks
CLC	8086	2
	8088	2
	80286	2
	80386	2
	80486	2
	Pentium–Core2	2

CLD Clear direction flag

11111100 O D I T S Z A P C
0

Example	Microprocessor	Clocks
CLD	8086	2
	8088	2
	80286	2
	80386	2
	80486	2
	Pentium–Core2	2

CLI Clear interrupt flag

11111010 O D I T S Z A P C
0

Example	Microprocessor	Clocks
CLI	8086	2
	8088	2
	80286	3
	80386	3
	80486	5
	Pentium–Core2	7

CLTS Clear task switched flag (CR0)

00001111 00000110

Example	Microprocessor	Clocks
CLTS	8086	—
	8088	—
	80286	2
	80386	5
	80486	7
	Pentium–Core2	10

CMC Complement carry flag

10011000 O D I T S Z A P C
*

Example	Microprocessor	Clocks
CMC	8086	2
	8088	2
	80286	2
	80386	2
	80486	2
	Pentium–Core2	2

CMOVcondition Conditional move

00001111 0100cccc oorrrmmm

Format	Examples	Microprocessor	Clocks
CMOVcc reg,mem	CMOVNZ AX,FROG CMOVC EAX,[EDI] CMOVNC BX,DATA1 CMOVP EBX,WAITING CMOVNE DI,[SI]	8086	—
		8088	—
		80286	—
		80386	—
		80486	—
		Pentium–Core2	—

Condition Codes	Mnemonic	Flag	Description
0000	CMOVO	O = 1	Move if overflow
0001	CMOVNO	O = 0	Move if no overflow
0010	CMOVB	C = 1	Move if below
0011	CMOVAE	C = 0	Move if above or equal
0100	CMOVE	Z = 1	Move if equal/zero
0101	CMOVNE	Z = 0	Move if not equal/zero
0110	CMOVBE	C = 1 + Z = 1	Move if below or equal
0111	CMOVA	C = 0 • Z = 0	Move if above
1000	CMOVS	S = 1	Move if sign
1001	CMOVNS	S = 0	Move if no sign
1010	CMOVP	P = 1	Move if parity
1011	CMOVNP	P = 0	Move if no parity
1100	CMOVL	S • O	Move if less than
1101	CMOVGE	S = 0	Move if greater than or equal
1110	CMOVLE	Z = 1 + S • O	Move if less than or equal
1111	CMOVG	Z = 0 + S = O	Move if greater than

CMP Compare

001110dw oorrrmmm disp

	O	D	I	T	S	Z	A	P	C
	*				*	*	*	*	*

Format	Examples	Microprocessor	Clocks
CMP reg,reg	CMP AX,BX CMP AL,BL CMP EAX,EBX CMP CX,SI CMP ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2

CMP mem,reg	CMP DATAY,AL CMP LIST,SI CMP DATA6[DI],CL CMP [EAX],CL CMP [EDX+4*ECX],EBX	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	5
		80486	2
		Pentium–Core2	1 or 2
CMP reg,mem	CMP BL,DATA2 CMP SI,LIST3 CMP CL,DATA2[DI] CMP CX,[EDI] CMP ESI,[ECX+200H]	8086	9 + ea
		8088	13 + ea
		80286	6
		80386	6
		80486	2
		Pentium–Core2	1 or 2
100000sw oo111mmm disp data Format	Examples	Microprocessor	Clocks
CMP reg,imm	CMP CX,3 CMP DI,1AH CMP DL,34H CMP EDX,1345H CMP CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 2
CMP mem,imm	CMP DATAS,3 CMP BYTE PTR[EDI],1AH CMP DADDY,34H CMP LIST,'A' CMP TOAD,1834H	8086	10 + ea
		8088	14 + ea
		80286	6
		80386	5
		80486	2
		Pentium–Core2	1 or 2

0001111w data

Format	Examples	Microprocessor	Clocks
CMP acc,imm	CMP AX,3 CMP AL,1AH CMP AH,34H CMP EAX,1345H CMP AL,'Y'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

CMPS Compare strings

1010011w

O D I T S Z A P C
* * * * * *

Format	Examples	Microprocessor	Clocks
CMPSB CMPSW CMPSD	CMPSB CMPSW CMPSD CMPSB DATA1,DATA2 REPE CMPSB REPNE CMPSW	8086	32
		8088	30
		80286	8
		80386	10
		80486	8
		Pentium–Core2	5

CMPXCHG Compare and exchange

00001111 1011000w 11rrrrrr

O D I T S Z A P C
* * * * * *

Format	Examples	Microprocessor	Clocks
CMPXCHG reg,reg	CMPXCHG EAX,EBX CMPXCHG ECX,EDX	8086	—
		8088	—
		80286	—
		80386	—
		80486	6
		Pentium–Core2	6

0001111w data

Format	Examples	Microprocessor	Clocks
CMPXCHG mem,reg	CMPXCHG DATAD,EAX CMPXCHG DATA2,EDI	8086	—
		8088	—
		80286	—
		80386	—
		80486	7
		Pentium–Core2	6

CMPXCHG8B Compare and exchange 8 bytes

00001111 11000111 oorrrmmm

O D I T S Z A P C
*

Format	Examples	Microprocessor	Clocks
CMPXCHG8B mem64	CMPXCHG8B DATA3	8086	—
		8088	—
		80286	—
		80386	—
		80486	—
		Pentium–Core2	10

CPUID CPU identification code

00001111 10100010

Example	Microprocessor	Clocks
CPUID	8086	—
	8088	—
	80286	—
	80386	—
	80486	—
	Pentium–Core2	14

CWD Convert word to doubleword (AX ⇒ DX:AX)

10011000

Example	Microprocessor	Clocks
CWD	8086	5
	8088	5
	80286	2
	80386	2
	80486	3
	Pentium–Core2	2

CWDE Convert word to extended doubleword (AX ⇒ EAX)

10011000

Example	Microprocessor	Clocks
CWDE	8086	—
	8088	—
	80286	—
	80386	3
	80486	3
	Pentium–Core2	3

DAA Decimal adjust AL after addition

00100111

O	D	I	T	S	Z	A	P	C
?				*	*	*	*	*

Example	Microprocessor	Clocks
DAA	8086	4
	8088	4
	80286	3
	80386	4
	80486	2
	Pentium–Core2	3

DAS Decimal adjust AL after subtraction

00101111

O	D	I	T	S	Z	A	P	C
?				*	*	*	*	*

Example	Microprocessor	Clocks
DAS	8086	4
	8088	4
	80286	3
	80386	4
	80486	2
	Pentium–Core2	3

DEC Decrement

1111111w oo001mmm disp

O	D	I	T	S	Z	A	P	C
*				*	*	*	*	

Format	Examples	Microprocessor	Clocks
DEC reg8	DEC BL DEC BH DEC CL DEC DH DEC AH	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3
DEC mem	DEC DATAY DEC LIST DEC DATA6[DI] DEC BYTE PTR [BX] DEC WORD PTR [EBX]	8086	15 + ea
		8088	23 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3

01001rrr

Format	Examples	Microprocessor	Clocks
DEC reg16 DEC reg32	DEC CX DEC DI DEC EDX DEC ECX DEC BP	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1

DIV Divide

1111011w oo110mmm disp

O D I T S Z A P C
? ? ? ? ? ? ? ? ?

Format	Examples	Microprocessor	Clocks
DIV reg	DIV BL DIV BH DIV ECX DIV DH DIV CX	8086	162
		8088	162
		80286	22
		80386	38
		80486	40
		Pentium–Core2	17–41
DIV mem	DIV DATAY DIV LIST DIV DATA6[DI] DIV BYTE PTR [BX] DIV WORD PTR [EBX]	8086	168
		8088	176
		80286	25
		80386	41
		80486	40
		Pentium–Core2	17–41

ENTER Create a stack frame			
11001000 data Format	Examples	Microprocessor	Clocks
ENTER imm,0	ENTER 4,0 ENTER 8,0 ENTER 100,0 ENTER 200,0 ENTER 1024,0	8086	—
		8088	—
		80286	11
		80386	10
		80486	14
		Pentium–Core2	11
ENTER imm,1	ENTER 4,1 ENTER 10,1	8086	—
		8088	—
		80286	12
		80386	15
		80486	17
		Pentium–Core2	15
ENTER imm,imm	ENTER 3,6 ENTER 100,3	8086	—
		8088	—
		80286	12
		80386	15
		80486	17
		Pentium–Core2	15 + 2n
ESC Escape (obsolete–see coprocessor)			

HLT Halt

11110100

Example	Microprocessor	Clocks
HLT	8086	2
	8088	2
	80286	2
	80386	5
	80486	4
	Pentium–Core2	varies

IDIV Integer (signed) division

1111011w oo111mmm disp

O D I T S Z A P C
? ? ? ? ? ? ?

Format	Examples	Microprocessor	Clocks
IDIV reg	IDIV BL IDIV BH IDIV ECX IDIV DH IDIV CX	8086	184
		8088	184
		80286	25
		80386	43
		80486	43
		Pentium–Core2	22–46
IDIV mem	IDIV DATAY IDIV LIST IDIV DATA6[DI] IDIV BYTE PTR [BX] IDIV WORD PTR [EBX]	8086	190
		8088	194
		80286	28
		80386	46
		80486	44
		Pentium–Core2	22–46

IMUL Integer (signed) multiplication

1111011w oo101mmm disp

O D I T S Z A P C
* ? ? ? ? *

Format	Examples	Microprocessor	Clocks
IMUL reg	IMUL BL IMUL CX IMUL ECX IMUL DH IMUL AL	8086	154
		8088	154
		80286	21
		80386	38
		80486	42
		Pentium–Core2	10–11
IMUL mem	IMUL DATAY IMUL LIST IMUL DATA6[DI] IMUL BYTE PTR [BX] IMUL WORD PTR [EBX]	8086	160
		8088	164
		80286	24
		80386	41
		80486	42
		Pentium–Core2	10–11

011010s1 oorrmmm disp data

Format	Examples	Microprocessor	Clocks
IMUL reg,imm	IMUL CX,16 IMUL DI,100 IMUL EDX,20	8086	—
		8088	—
		80286	21
		80386	38
		80486	42
		Pentium–Core2	10
IMUL reg,reg,imm	IMUL DX,AX,2 IMUL CX,DX,3 IMUL BX,AX,33	8086	—
		8088	—
		80286	21
		80386	38
		80486	42
		Pentium–Core2	10

IMUL reg,mem,imm	IMUL CX,DATAY,99	8086	—
		8088	—
		80286	24
		80386	38
		80486	42
		Pentium–Core2	10
00001111 10101111 oorrmmm disp Format	Examples	Microprocessor	Clocks
IMUL reg,reg	IMUL CX,DX IMUL DI,BX IMUL EDX,EBX	8086	—
		8088	—
		80286	—
		80386	38
		80486	42
		Pentium–Core2	10
IMUL reg,mem	IMUL DX,DATAY IMUL CX,LIST IMUL ECX,DATA6[DI]	8086	—
		8088	—
		80286	—
		80386	41
		80486	42
		Pentium–Core2	10
IN	Input data from port		
1110010w port# Format	Examples	Microprocessor	Clocks
IN acc,pt	IN AL,12H IN AX,12H IN AL,0FFH IN AX,0A0H IN EAX,10H	8086	10
		8088	14
		80286	5
		80386	12
		80486	14
		Pentium–Core2	7

1110110w Format	Examples	Microprocessor	Clocks
IN acc,DX	IN AL,DX IN AX,DX IN EAX,DX	8086	8
		8088	12
		80286	5
		80386	13
		80486	14
		Pentium–Core2	7

INC Increment

1111111w oo000mmm disp

O D I T S Z A P C
* * * * *

Format	Examples	Microprocessor	Clocks
INC reg8	INC BL INC BH INC AL INC AH INC DH	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3
INC mem	INC DATA3 INC LIST INC COUNT INC BYTE PTR [DI] INC WORD PTR [ECX]	8086	15 + ea
		8088	23 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3
INC reg16 INC reg32	INC CX INC DX INC BP INC ECX INC ESP	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1

INS Input string from port

0110110w

Format	Examples	Microprocessor	Clocks
INSB INSW INSD	INSB INSW INSD INS DATA2 REP INSB	8086	—
		8088	—
		80286	5
		80386	15
		80486	17
		Pentium–Core2	9

INT Interrupt

11001101 type

Format	Examples	Microprocessor	Clocks
INT type	INT12H INT15H INT 21H INT 2FH INT 10H	8086	51
		8088	71
		80286	23
		80386	37
		80486	30
		Pentium–Core2	16–82

INT 3 Interrupt 3

11001100

Example	Microprocessor	Clocks
INT 3	8086	52
	8088	72
	80286	23
	80386	33
	80486	26
	Pentium–Core2	13–56

INTO Interrupt on overflow

11001110

Example	Microprocessor	Clocks
INTO	8086	53
	8088	73
	80286	24
	80386	35
	80486	28
	Pentium–Core2	13–56

INVD Invalidate data cache

00001111 00001000

Example	Microprocessor	Clocks
INTVD	8086	—
	8088	—
	80286	—
	80386	—
	80486	4
	Pentium–Core2	15

IRET/IRETD Return from interrupt

11001101 data

O	D	I	T	S	Z	A	P	C
*	*	*	*	*	*	*	*	*

Format	Examples	Microprocessor	Clocks
IRET IRETD	IRET IRETD IRET 100	8086	32
		8088	44
		80286	17
		80386	22
		80486	15
		Pentium–Core2	8–27

Jcondition Conditional jump

0111cccc disp

Format	Examples	Microprocessor	Clocks
Jcnd label (8-bit disp)	JA ABOVE JB BELOW JG GREATER JE EQUAL JZ ZERO	8086	16/4
		8088	16/4
		80286	7/3
		80386	7/3
		80486	3/1
		Pentium–Core2	1

00001111 1000cccc disp

Format	Examples	Microprocessor	Clocks
Jcnd label (16-bit disp)	JNE NOT_MORE JLE LESS_OR_SO	8086	—
		8088	—
		80286	—
		80386	7/3
		80486	3/1
		Pentium–Core2	1

Condition Codes	Mnemonic	Flag	Description
0000	JO	O = 1	Jump if overflow
0001	JNO	O = 0	Jump if no overflow
0010	JB/NAE	C = 1	Jump if below
0011	JAE/JNB	C = 0	Jump if above or equal
0100	JE/JZ	Z = 1	Jump if equal/zero
0101	JNE/JNZ	Z = 0	Jump if not equal/zero
0110	JBE/JNA	C = 1 + Z = 1	Jump if below or equal
0111	JA/JNBE	C = 0 • Z = 0	Jump if above
1000	JS	S = 1	Jump if sign
1001	JNS	S = 0	Jump if no sign
1010	JP/JPE	P = 1	Jump if parity
1011	JNP/JPO	P = 0	Jump if no parity
1100	JL/JNGE	S • O	Jump if less than
1101	JGE/JNL	S = 0	Jump if greater than or equal
1110	JLE/JNG	Z = 1 + S • O	Jump if less than or equal
1111	JG/JNLE	Z = 0 + S = O	Jump if greater than

JCXZ/JECXZ Jump if CX (ECX) equals zero

11100011 Format	Examples	Microprocessor	Clocks
JCXZ label JECXZ label	JCXZ ABOVE JCXZ BELOW JECXZ GREATER JECXZ EQUAL JCXZ NEXT	8086	18/6
		8088	18/6
		80286	8/4
		80386	9/5
		80486	8/5
		Pentium–Core2	6/5

JMP Jump

11101011 disp Format	Examples	Microprocessor	Clocks
JMP label (short)	JMP SHORT UP JMP SHORT DOWN JMP SHORT OVER JMP SHORT CIRCUIT JMP SHORT JOKE	8086	15
		8088	15
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1

11101001 disp Format	Examples	Microprocessor	Clocks
JMP label (near)	JMP VERS JMP FROG JMP UNDER JMP NEAR PTR OVER	8086	15
		8088	15
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1

11101010 disp Format	Examples	Microprocessor	Clocks
JMP label (far)	JMP NOT_MORE JMP UNDER JMP AGAIN JMP FAR PTR THERE	8086	15
		8088	15
		80286	11
		80386	12
		80486	17
		Pentium–Core2	3
11111111 oo100mmm Format	Examples	Microprocessor	Clocks
JMP reg (near)	JMP AX JMP EAX JMP CX JMP DX	8086	11
		8088	11
		80286	7
		80386	7
		80486	3
		Pentium–Core2	2
JMP mem (near)	JMP VERS JMP FROG JMP CS:UNDER JMP DATA1[DI+2]	8086	18 + ea
		8088	18 + ea
		80286	11
		80386	10
		80486	5
		Pentium–Core2	4
11111111 oo101mmm Format	Examples	Microprocessor	Clocks
JMP mem (far)	JMP WAY_OFF JMP TABLE JMP UP JMP OUT_OF_HERE	8086	24 + ea
		8088	24 + ea
		80286	15
		80386	12
		80486	13
		Pentium–Core2	4

LAHF Load AH from flags

10011111

Example	Microprocessor	Clocks
LAHF	8086	4
	8088	4
	80286	2
	80386	2
	80486	3
	Pentium–Core2	2

LAR Load access rights byte

00001111 00000010 oorrrmmm disp

O D I T S Z A P C
*

Format	Examples	Microprocessor	Clocks
LAR reg,reg	LAR AX,BX LAR CX,DX LAR ECX,EDX	8086	—
		8088	—
		80286	14
		80386	15
		80486	11
		Pentium–Core2	8
LAR reg,mem	LAR CX,DATA1 LAR AX,LIST3 LAR ECX,TOAD	8086	—
		8088	—
		80286	16
		80386	16
		80486	11
		Pentium–Core2	8

LDS Load far pointer to DS and register

11000101 oorrrmmm

Format	Examples	Microprocessor	Clocks
LDS reg,mem	LDS DI,DATA3 LDS SI,LIST2 LDS BX,ARRAY_PTR LDS CX,PNTR	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	6
		Pentium–Core2	4

LEA Load effective address

10001101 oorrrmmm disp

Format	Examples	Microprocessor	Clocks
LEA reg,mem	LEA DI,DATA3 LEA SI,LIST2 LEA BX,ARRAY_PTR LEA CX,PNTR	8086	2 + ea
		8088	2 + ea
		80286	3
		80386	2
		80486	2
		Pentium–Core2	1

LEAVE Leave high-level procedure

11001001

Example	Microprocessor	Clocks
LEAVE	8086	—
	8088	—
	80286	5
	80386	4
	80486	5
	Pentium–Core2	3

LES Load far pointer to ES and register

11000100 oorrrmmm

Format	Examples	Microprocessor	Clocks
LES reg,mem	LES DI,DATA3 LES SI,LIST2 LES BX,ARRAY_PTR LES CX,PNTR	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	6
		Pentium–Core2	4

LFS Load far pointer to FS and register

00001111 10110100 oorrrmmm disp

Format	Examples	Microprocessor	Clocks
LFS reg,mem	LFS DI,DATA3 LFS SI,LIST2 LFS BX,ARRAY_PTR LFS CX,PNTR	8086	—
		8088	—
		80286	—
		80386	7
		80486	6
		Pentium–Core2	4

LGDT Load global descriptor table

00001111 00000001 oo010mmm disp

Format	Examples	Microprocessor	Clocks
LGDT mem64	LGDT DESCRIP LGDT TABLED	8086	—
		8088	—
		80286	11
		80386	11
		80486	11
		Pentium–Core2	6

LGS Load far pointer to GS and register

00001111 10110101 oorrrmmm disp

Format	Examples	Microprocessor	Clocks
LGS reg,mem	LGS DI,DATA3 LGS SI,LIST2 LGS BX,ARRAY_PTR LGS CX,PNTR	8086	—
		8088	—
		80286	—
		80386	7
		80486	6
		Pentium–Core2	4

LIDT Load interrupt descriptor table

00001111 00000001 oo011mmm disp

Format	Examples	Microprocessor	Clocks
LIDT mem64	LIDT DATA3 LIDT LIST2	8086	—
		8088	—
		80286	12
		80386	11
		80486	11
		Pentium–Core2	6

LLDT Load local descriptor table

00001111 00000000 oo010mmm disp

Format	Examples	Microprocessor	Clocks
LLDT reg	LLDT BX LLDT DX LLDT CX	8086	—
		8088	—
		80286	17
		80386	20
		80486	11
		Pentium–Core2	9

LLDT mem	LLDT DATA1 LLDT LIST3 LLDT TOAD	8086	—
		8088	—
		80286	19
		80386	24
		80486	11
		Pentium–Core2	9

LMSW Load machine status word (80286 only)

00001111 00000001 oo110mmm disp

Format	Examples	Microprocessor	Clocks
LMSW reg	LMSW BX LMSW DX LMSW CX	8086	—
		8088	—
		80286	3
		80386	10
		80486	2
		Pentium–Core2	8
LMSW mem	LMSW DATA1 LMSW LIST3 LMSW TOAD	8086	—
		8088	—
		80286	6
		80386	13
		80486	3
		Pentium–Core2	8

LOCK Lock the bus

11110000

Format	Examples	Microprocessor	Clocks
LOCK:inst	LOCK:XCHG AX,BX LOCK:ADD AL,3	8086	2
		8088	3
		80286	0
		80386	0
		80486	1
		Pentium–Core2	1

LODS Load string operand

1010110w

Format	Examples	Microprocessor	Clocks
LODSB LODSW LODSD	LODSB LODSW LODSD LODS DATA3	8086	12
		8088	15
		80286	5
		80386	5
		80486	5
		Pentium–Core2	2

LOOP/LOOPD Loop until CX = 0 or ECX = 0

11100010 disp

Format	Examples	Microprocessor	Clocks
LOOP label LOOPD label	LOOP NEXT LOOP BACK LOOPD LOOPS	8086	17/5
		8088	17/5
		80286	8/4
		80386	11
		80486	7/6
		Pentium–Core2	5/6

LOOPE/LOOPED Loop while equal

11100001 disp

Format	Examples	Microprocessor	Clocks
LOOPE label LOOPED label LOOPZ label LOOPZD label	LOOPE AGAIN LOOPED UNTIL LOOPZ ZORRO LOOPZD WOW	8086	18/6
		8088	18/6
		80286	8/4
		80386	11
		80486	9/6
		Pentium–Core2	7/8

LOOPNE/LOOPNED Loop while not equal

11100000 disp

Format	Examples	Microprocessor	Clocks
LOOPNE label LOOPNED label LOOPNZ label LOOPNZD label	LOOPNE FORWARD LOOPNED UPS LOOPNZ TRY_AGAIN LOOPNZD WOO	8086	19/5
		8088	19/5
		80286	8/4
		80386	11
		80486	9/6
		Pentium–Core2	7/8

LSL Load segment limit

00001111 00000011 oorrrmmm disp

O D I T S Z A P C
*

Format	Examples	Microprocessor	Clocks
LSL reg,reg	LSL AX,BX LSL CX,BX LSL EDX,EAX	8086	—
		8088	—
		80286	14
		80386	25
		80486	10
		Pentium–Core2	8

LSL reg,mem	LSL AX,LIMIT LSL EAX,NUM	8086	—
		8088	—
		80286	16
		80386	26
		80486	10
		Pentium–Core2	8

LSS Load far pointer to SS and register

00001111 10110010 oorrrmmm disp

Format	Examples	Microprocessor	Clocks
LSS reg,mem	LSS DI,DATA1 LSS SP,STACK_TOP LSS CX,ARRAY	8086	—
		8088	—
		80286	—
		80386	7
		80486	6
		Pentium–Core2	4

LTR Load task register

00001111 00000000 oo001mmm disp

Format	Examples	Microprocessor	Clocks
LTR reg	LTR AX LTR CX LTR DX	8086	—
		8088	—
		80286	17
		80386	23
		80486	20
		Pentium–Core2	10

LTR mem16	LTR TASK LTR NUM	8086	—
		8088	—
		80286	19
		80386	27
		80486	20
		Pentium–Core2	10

MOV Move data

100010dw oorrrmmm disp

Format	Examples	Microprocessor	Clocks
MOV reg,reg	MOV CL,CH MOV BH,CL MOV CX,DX MOV EAX,EBP MOV ESP,ESI	8086	2
		8088	2
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1
MOV mem,reg	MOV DATA7,DL MOV NUMB,CX MOV TEMP,EBX MOV [ECX],BL MOV [DI],DH	8086	9 + ea
		8088	13 + ea
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1
MOV reg,mem	MOV DL,DATA8 MOV DX,NUMB MOV EBX,TEMP+3 MOV CH,TEMP[EDI] MOV CL,DATA2	8086	10 + ea
		8088	12 + ea
		80286	5
		80386	4
		80486	1
		Pentium–Core2	1

1100011w oo000mmm disp data Format	Examples	Microprocessor	Clocks
MOV mem,imm	MOV DATAF,23H MOV LIST,12H MOV BYTE PTR [DI],2 MOV NUMB,234H MOV DWORD PTR[ECX],1	8086	10 + ea
		8088	14 + ea
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1
1011wrrr data Format	Examples	Microprocessor	Clocks
MOV reg,imm	MOV BX,22H MOV CX,12H MOV CL,2 MOV ECX,123456H MOV DI,100	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1
101000dw disp Format	Examples	Microprocessor	Clocks
MOV mem,acc	MOV DATAF,AL MOV LIST,AX MOV NUMB,EAX	8086	10
		8088	14
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1
MOV acc,mem	MOV AL,DATAE MOV AX,LIST MOV EAX,LUTE	8086	10
		8088	14
		80286	5
		80386	4
		80486	1
		Pentium–Core2	1

100011d0 oosssmmm disp Format	Examples	Microprocessor	Clocks
MOV seg,reg	MOV SS,AX MOV DS,DX MOV ES,CX MOV FS,BX MOV GS,AX	8086	2
		8088	2
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1
MOV seg,mem	MOV SS,STACK_TOP MOV DS,DATAS MOV ES,TEMP1	8086	8 + ea
		8088	12 + ea
		80286	2
		80386	2
		80486	1
		Pentium–Core2	2 or 3
MOV reg,seg	MOV BX,DS MOV CX,FS MOV CX,ES	8086	2
		8088	2
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1
MOV mem,seg	MOV DATA2,CS MOV TEMP,DS MOV NUMB1,SS MOV TEMP2,GS	8086	9 + ea
		8088	13 + ea
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

00001111 001000d0 11rrrmmm Format	Examples	Microprocessor	Clocks
MOV reg,cr	MOV EBX,CR0 MOV ECX,CR2 MOV EBX,CR3	8086	—
		8088	—
		80286	—
		80386	6
		80486	4
		Pentium–Core2	4
MOV cr,reg	MOV CR0,EAX MOV CR1,EBX MOV CR3,EDX	8086	—
		8088	—
		80286	—
		80386	10
		80486	4
		Pentium–Core2	12–46
00001111 001000d1 11rrrmmm Format	Examples	Microprocessor	Clocks
MOV reg,dr	MOV EBX,DR6 MOV ECX,DR7 MOV EBX,DR1	8086	—
		8088	—
		80286	—
		80386	22
		80486	10
		Pentium–Core2	11
MOV dr,reg	MOV DR0,EAX MOV DR1,EBX MOV DR3,EDX	8086	—
		8088	—
		80286	—
		80386	22
		80486	11
		Pentium–Core2	11

00001111 001001d0 11rrrmmm Format	Examples	Microprocessor	Clocks
MOV reg,tr	MOV EBX,TR6 MOV ECX,TR7	8086	—
		8088	—
		80286	—
		80386	12
		80486	4
		Pentium–Core2	11
MOV tr,reg	MOV TR6,EAX MOV TR7,EBX	8086	—
		8088	—
		80286	—
		80386	12
		80486	6
		Pentium–Core2	11

MOVS Move string data

1010010w Format	Examples	Microprocessor	Clocks
MOVSB MOVSW MOVSD	MOVSB MOVSW MOVSD MOVS DATA1,DATA2	8086	18
		8088	26
		80286	5
		80386	7
		80486	7
		Pentium–Core2	4

MOVSX Move with sign extend			
00001111 1011111w oorrrmmm disp			
Format	Examples	Microprocessor	Clocks
MOVSX reg,reg	MOVSX BX,AL MOVSX EAX,DX	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	3
MOVSX reg,mem	MOVSX AX,DATA34 MOVSX EAX,NUMB	8086	—
		8088	—
		80286	—
		80386	6
		80486	3
		Pentium–Core2	3

MOVZX Move with zero extend			
00001111 1011011w oorrrmmm disp			
Format	Examples	Microprocessor	Clocks
MOVZX reg,reg	MOVZX BX,AL MOVZX EAX,DX	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	3
MOVZX reg,mem	MOVZX AX,DATA34 MOVZX EAX,NUMB	8086	—
		8088	—
		80286	—
		80386	6
		80486	3
		Pentium–Core2	3

MUL Multiply

1111011w oo100mmm disp

O D I T S Z A P C
* ? ? ? ? *

Format	Examples	Microprocessor	Clocks
MUL reg	MUL BL MUL CX MUL EDX	8086	118
		8088	143
		80286	21
		80386	38
		80486	42
		Pentium–Core2	10 or 11
MUL mem	MUL DATA9 MUL WORD PTR [ESI]	8086	139
		8088	143
		80286	24
		80386	41
		80486	42
		Pentium–Core2	11

NEG Negate

1111011w oo011mmm disp

O D I T S Z A P C
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Format	Examples	Microprocessor	Clocks
NEG reg	NEG BL NEG CX NEG EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3

NEG mem	NEG DATA9 NEG WORD PTR [ESI]	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3

NOP No operation

10010000

Example	Microprocessor	Clocks
NOP	8086	3
	8088	3
	80286	3
	80386	3
	80486	3
	Pentium–Core2	1

NOT One's complement

1111011w oo010mmm disp

Format	Examples	Microprocessor	Clocks
NOT reg	NOT BL NOT CX NOT EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 3
NOT mem	NOT DATA9 NOT WORD PTR [ESI]	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3

OR Inclusive-OR

000010dw oorrrmmm disp

O D I T S Z A P C
0 * * ? * 0

Format	Examples	Microprocessor	Clocks
OR reg,reg	OR AX,BX OR AL,BL OR EAX,EBX OR CX,SI OR ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2
OR mem,reg	OR DATAY,AL OR LIST,SI OR DATA2[DI],CL OR [EAX],BL OR [EBX+2*ECX],EDX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
OR reg,mem	OR BL,DATA1 OR SI,LIST1 OR CL,DATA2[SI] OR CX,[ESI] OR ESI,[2*ECX]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	6
		80486	2
		Pentium–Core2	1 or 3

100000sw oo001mmm disp data

Format	Examples	Microprocessor	Clocks
OR reg,imm	OR CX,3 OR DI,1AH OR DL,34H OR EDX,1345H OR CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3

OR mem,imm	OR DATAS,3 OR BYTE PTR[EDI],1AH OR DADDY,34H OR LIST,'A' OR TOAD,1834H	8086	17 + ea
		8088	25 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
0000110w data Format	Examples	Microprocessor	Clocks
OR acc,imm	OR AX,3 OR AL,1AH OR AH,34H OR EAX,1345H OR AL,'Y'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

OUT Output data to port

1110011w port# Format	Examples	Microprocessor	Clocks
OUT pt,acc	OUT 12H,AL OUT 12H,AX OUT 0FFH,AL OUT 0A0H,AX OUT 10H,EAX	8086	10
		8088	14
		80286	3
		80386	10
		80486	10
		Pentium–Core2	12–26
1110111w Format	Examples	Microprocessor	Clocks
OUT DX,acc	OUT DX,AL OUT DX,AX OUT DX,EAX	8086	8
		8088	12
		80286	3
		80386	11
		80486	10
		Pentium–Core2	12–26

OUTS	Output string to port		
0110111w Format	Examples	Microprocessor	Clocks
OUTSB OUTSW OUTSD	OUTSB OUTSW OUTSD OUTS DATA2 REP OUTSB	8086	—
		8088	—
		80286	5
		80386	14
		80486	10
		Pentium–Core2	13–27

POP	Pop data from stack		
01011rrr Format	Examples	Microprocessor	Clocks
POP reg	POP CX POP AX POP EDI	8086	8
		8088	12
		80286	5
		80386	4
		80486	1
		Pentium–Core2	1
10001111 oo000mmm disp Format	Examples	Microprocessor	Clocks
POP mem	POP DATA1 POP LISTS POP NUMBS	8086	17 + ea
		8088	25 + ea
		80286	5
		80386	5
		80486	4
		Pentium–Core2	3

00sss111 Format	Examples	Microprocessor	Clocks
POP seg	POP DS POP ES POP SS	8086	8
		8088	12
		80286	5
		80386	7
		80486	3
		Pentium–Core2	3
00001111 10sss001 Format	Examples	Microprocessor	Clocks
POP seg	POP FS POP GS	8086	—
		8088	—
		80286	—
		80386	7
		80486	3
		Pentium–Core2	3

POPA/POPAD Pop all registers from stack

01100001 Example	Microprocessor	Clocks
POPA POPAD	8086	—
	8088	—
	80286	19
	80386	24
	80486	9
	Pentium–Core2	5

POPF/POPFD Pop flags from stack

10010000

O D I T S Z A P C
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Example	Microprocessor	Clocks
POPF POPFD	8086	8
	8088	12
	80286	5
	80386	5
	80486	6
	Pentium–Core2	4 or 6

PUSH Push data onto stack

01010rrr

Format	Examples	Microprocessor	Clocks
PUSH reg	PUSH CX PUSH AX PUSH EDI	8086	11
		8088	15
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

11111111 oo110mmm disp

Format	Examples	Microprocessor	Clocks
PUSH mem	PUSH DATA1 PUSH LISTS PUSH NUMBS	8086	16 + ea
		8088	24 + ea
		80286	5
		80386	5
		80486	4
		Pentium–Core2	1 or 2

00ss110 Format	Examples	Microprocessor	Clocks
PUSH seg	PUSH ES PUSH CS PUSH DS	8086	10
		8088	14
		80286	3
		80386	2
		80486	3
		Pentium–Core2	1
00001111 10sss000 Format	Examples	Microprocessor	Clocks
PUSH seg	PUSH FS PUSH GS	8086	—
		8088	—
		80286	—
		80386	2
		80486	3
		Pentium–Core2	1
011010s0 data Format	Examples	Microprocessor	Clocks
PUSH imm	PUSH 2000H PUSH 53220 PUSHW 10H PUSH ‘;’ PUSHD 100000H	8086	—
		8088	—
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

PUSHA/PUSHAD Push all registers onto stack

01100000

Example	Microprocessor	Clocks
PUSHA PUSHAD	8086	—
	8088	—
	80286	17
	80386	18
	80486	11
	Pentium–Core2	5

PUSHF/PUSHFD Push flags onto stack

10011100

Example	Microprocessor	Clocks
PUSHF PUSHFD	8086	10
	8088	14
	80286	3
	80386	4
	80486	3
	Pentium–Core2	3 or 4

RCL/RCR/ROL/ROR Rotate

1101000w ooTTTmmm disp

O	D	I	T	S	Z	A	P	C
*								*

TTT = 000 = ROL, TTT = 001 = ROR, TTT = 010 = RCL, and TTT = 011 = RCR

Format	Examples	Microprocessor	Clocks
ROL reg,1 ROR reg,1	ROL CL,1 ROL DX,1 ROR CH,1 ROR SI,1	8086	2
		8088	2
		80286	2
		80386	3
		80486	3
		Pentium–Core2	1 or 3

RCL reg,1 RCR reg,1	RCL CL,1 RCL SI,1 RCR AH,1 RCR EBX,1	8086	2
		8088	2
		80286	2
		80386	9
		80486	3
		Pentium–Core2	1 or 3
ROL mem,1 ROR mem,1	ROL DATAY,1 ROL LIST,1 ROR DATA2[DI],1 ROR BYTE PTR [EAX],1	8086	15 + ea
		8088	23 + ea
		80286	7
		80386	7
		80486	4
		Pentium–Core2	1 or 3
RCL mem,1 RCR mem,1	RCL DATA1,1 RCL LIST,1 RCR DATA2[SI],1 RCR WORD PTR [ESI],1	8086	15 + ea
		8088	23 + ea
		80286	7
		80386	10
		80486	4
		Pentium–Core2	1 or 3
1101001w ooTTTmmm disp Format	Examples	Microprocessor	Clocks
ROL reg,CL ROR reg,CL	ROL CH,CL ROL DX,CL ROR AL,CL ROR ESI,CL	8086	8 + 4n
		8088	8 + 4n
		80286	5 + n
		80386	3
		80486	3
		Pentium–Core2	4

RCL reg,CL RCR reg,CL	RCL CH,CL RCL SI,CL RCR AH,CL RCR EBX,CL	8086	8 + 4n
		8088	8 + 4n
		80286	5 + n
		80386	9
		80486	3
		Pentium–Core2	7–27
ROL mem,CL ROR mem,CL	ROL DATAY,CL ROL LIST,CL ROR DATA2[DI],CL ROR BYTE PTR [EAX],CL	8086	20 + 4n
		8088	28 + 4n
		80286	8 + n
		80386	7
		80486	4
		Pentium–Core2	4
RCL mem,CL RCR mem,CL	RCL DATA1,CL RCL LIST,CL RCR DATA2[SI],CL RCR WORD PTR [ESI],CL	8086	20 + 4n
		8088	28 + 4n
		80286	8 + n
		80386	10
		80486	9
		Pentium–Core2	9–26
1100000w ooTTTmmm disp data Format	Examples	Microprocessor	Clocks
ROL reg,imm ROR reg,imm	ROL CH,4 ROL DX,5 ROR AL,2 ROR ESI,14	8086	—
		8088	—
		80286	5 + n
		80386	3
		80486	2
		Pentium–Core2	1 or 3

RCL reg,imm RCR reg,imm	RCL CL,2 RCL SI,12 RCR AH,5 RCR EBX,18	8086	—
		8088	—
		80286	5 + n
		80386	9
		80486	8
		Pentium–Core2	8–27
ROL mem,imm ROR mem,imm	ROL DATAY,4 ROL LIST,3 ROR DATA2[DI],7 ROR BYTE PTR [EAX],11	8086	—
		8088	—
		80286	8 + n
		80386	7
		80486	4
		Pentium–Core2	1 or 3
RCL mem,imm RCR mem,imm	RCL DATA1,5 RCL LIST,3 RCR DATA2[SI],9 RCR WORD PTR [ESI],8	8086	—
		8088	—
		80286	8 + n
		80386	10
		80486	9
		Pentium–Core2	8–27

RDMSR Read model specific register

00001111 00110010

Example	Microprocessor	Clocks
RDMSR	8086	—
	8088	—
	80286	—
	80386	—
	80486	—
	Pentium–Core2	20–24

REP Repeat prefix			
11110011 1010010w			
Format	Examples	Microprocessor	Clocks
REP MOVS	REP MOVSB REP MOVSW REP MOVSD REP MOVS DATA1,DATA2	8086	9 + 17n
		8088	9 + 25n
		80286	5 + 4n
		80386	8 + 4n
		80486	12 + 3n
		Pentium–Core2	13 + n
11110011 1010101w			
Format	Examples	Microprocessor	Clocks
REP STOS	REP STOSB REP STOSW REP STOSD REP STOS ARRAY	8086	9 + 10n
		8088	9 + 14n
		80286	4 + 3n
		80386	5 + 5n
		80486	7 + 4n
		Pentium–Core2	9 + n
11110011 0110110w			
Format	Examples	Microprocessor	Clocks
REP INS	REP INSB REP INSW REP INSD REP INS ARRAY	8086	—
		8088	—
		80286	5 + 4n
		80386	12 + 5n
		80486	17 + 5n
		Pentium–Core2	25 + 3n

11110011 0110111w Format	Examples	Microprocessor	Clocks
REP OUTS	REP OUTSB REP OUTSW REP OUTSD REP OUTS ARRAY	8086	—
		8088	—
		80286	5 + 4n
		80386	12 + 5n
		80486	17 + 5n
		Pentium–Core2	25 + 4n

REPE/REPNE Repeat conditional

11110011 1010011w Format	Examples	Microprocessor	Clocks
REPE CMPS	REPE CMPSB REPE CMPSW REPE CMPSD REPE CMPS DATA1,DATA2	8086	9 + 22n
		8088	9 + 30n
		80286	5 + 9n
		80386	5 + 9n
		80486	7 + 7n
		Pentium–Core2	9 + 4n
11110011 1010111w Format	Examples	Microprocessor	Clocks
REPE SCAS	REPE SCASB REPE SCASW REPE SCASD REPE SCAS ARRAY	8086	9 + 15n
		8088	9 + 19n
		80286	5 + 8n
		80386	5 + 8n
		80486	7 + 5n
		Pentium–Core2	9 + 4n

11110010 1010011w Format	Examples	Microprocessor	Clocks
REPNE CMPS	REPNE CMPSB REPNE CMPSW REPNE CMPSD REPNE CMPS ARRAY,LIST	8086	9 + 22n
		8088	9 + 30n
		80286	5 + 9n
		80386	5 + 9n
		80486	7 + 7n
		Pentium–Core2	8 + 4n
11110010 101011w Format	Examples	Microprocessor	Clocks
REPNE SCAS	REPNE SCASB REPNE SCASW REPNE SCASD REPNE SCAS ARRAY	8086	9 + 15n
		8088	9 + 19N
		80286	5 + 8n
		80386	5 + 8n
		80486	7 + 5n
		Pentium–Core2	9 + 4n

RET Return from procedure

11000011 Example	Microprocessor	Clocks
RET (near)	8086	16
	8088	20
	80286	11
	80386	10
	80486	5
	Pentium–Core2	2

11000010 data Format	Examples	Microprocessor	Clocks
RET imm (near)	RET 4 RET 100H	8086	20
		8088	24
		80286	11
		80386	10
		80486	5
		Pentium–Core2	3
11001011 Example		Microprocessor	Clocks
RET (far)		8086	26
		8088	34
		80286	15
		80386	18
		80486	13
		Pentium–Core2	4–23
11001010 data Format	Examples	Microprocessor	Clocks
RET imm (far)	RET 4 RET 100H	8086	25
		8088	33
		80286	11
		80386	10
		80486	5
		Pentium–Core2	4–23

RSM Resume from system management mode

00001111 10101010

O D I T S Z A P C
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Example	Microprocessor	Clocks
RSM	8086	—
	8088	—
	80286	—
	80386	—
	80486	—
	Pentium–Core2	83

SAHF Store AH into flags

10011110

O D I T S Z A P C
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Example	Microprocessor	Clocks
SAHF	8086	4
	8088	4
	80286	2
	80386	3
	80486	2
	Pentium–Core2	2

SAL/SAR/SHL/SHR Shift

1101000w ooTTTmmm disp

O D I T S Z A P C
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TTT = 100 = SHL/SAL , TTT = 101 = SHR, and TTT = 111 = SAR

Format	Examples	Microprocessor	Clocks
SAL reg,1 SHL reg,1 SHR reg,1 SAR reg,1	SAL CL,1 SHL DX,1 SAR CH,1 SHR SI,1	8086	2
		8088	2
		80286	2
		80386	3
		80486	3
		Pentium–Core2	1 or 3

SAL mem,1 SHL mem,1 SHR mem,1 SAR mem,1	SAL DATA1,1 SHL BYTE PTR [DI],1 SAR NUMB,1 SHR WORD PTR[EDI],1	8086	15 + ea
		8088	23 + ea
		80286	7
		80386	7
		80486	4
		Pentium–Core2	1 or 3
1101001w ooTTTmmm disp Format	Examples	Microprocessor	Clocks
SAL reg,CL SHL reg,CL SAR reg,CL SHR reg,CL	SAL CH,CL SHL DX,CL SAR AL,CL SHR ESI,CL	8086	8 + 4n
		8088	8 + 4n
		80286	5 + n
		80386	3
		80486	3
		Pentium–Core2	4
SAL mem,CL SHL mem,CL SAR mem,CL SHR mem,CL	SAL DATAU,CL SHL BYTE PTR [ESI],CL SAR NUMB,CL SHR TEMP,CL	8086	20 + 4n
		8088	28 + 4n
		80286	8 + n
		80386	7
		80486	4
		Pentium–Core2	4
1100000w ooTTTmmm disp data Format	Examples	Microprocessor	Clocks
SAL reg,imm SHL reg,imm SAR reg,imm SHR reg,imm	SAL CH,4 SHL DX,10 SAR AL,2 SHR ESI,23	8086	—
		8088	—
		80286	5 + n
		80386	3
		80486	2
		Pentium–Core2	1 or 3

SAL mem,imm SHL mem,imm SAR mem,imm SHR mem,imm	SAL DATAU,3 SHL BYTE PTR [ESI],15 SAR NUMB,3 SHR TEMP,5	8086	—
		8088	—
		80286	8 + n
		80386	7
		80486	4
		Pentium–Core2	1 or 3

SBB Subtract with borrow

000110dw oorrrmmm disp

O D I T S Z A P C
* * * * * *

Format	Examples	Microprocessor	Clocks
SBB reg,reg	SBB CL,DL SBB AX,DX SBB CH,CL SBB EAX,EBX SBB ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2
SBB mem,reg	SBB DATAJ,CL SBB BYTES,CX SBB NUMBS,ECX SBB [EAX],CX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3
SBB reg,mem	SBB CL,DATAL SBB CX,BYTES SBB ECX,NUMBS SBB DX,[EBX+EDI]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	7
		80486	2
		Pentium–Core2	1 or 2

100000sw oo011mmm disp data

Format	Examples	Microprocessor	Clocks
SBB reg,imm	SBB CX,3 SBB DI,1AH SBB DL,34H SBB EDX,1345H SBB CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3
SBB mem,imm	SBB DATAS,3 SBB BYTE PTR[EDI],1AH SBB DADDY,34H SBB LIST,'A' SBB TOAD,1834H	8086	17 + ea
		8088	25 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3

0001110w data

Format	Examples	Microprocessor	Clocks
SBB acc,imm	SBB AX,3 SBB AL,1AH SBB AH,34H SBB EAX,1345H SBB AL,'Y'	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

SCAS Scan string

1010111w

O	D	I	T	S	Z	A	P	C
*				*	*	*	*	*

Format	Examples	Microprocessor	Clocks
SCASB SCASW SCASD	SCASB SCASW SCASD SCAS DATAF REP SCASB	8086	15
		8088	19
		80286	7
		80386	7
		80486	6
		Pentium–Core2	4

SETcondition Conditional set

00001111 1001cccc oo000mmm

Format	Examples	Microprocessor	Clocks
SETcnd reg8	SETA BL SETB CH SETG DL SETE BH SETZ AL	8086	—
		8088	—
		80286	—
		80386	4
		80486	3
		Pentium–Core2	1 or 2
SETcnd mem8	SETE DATAK SETAE LESS_OR_SO	8086	—
		8088	—
		80286	—
		80386	5
		80486	3
		Pentium–Core2	1 or 2

Condition Codes	Mnemonic	Flag	Description
0000	SETO	O = 1	Set if overflow
0001	SETNO	O = 0	Set if no overflow
0010	SETB/SETAE	C = 1	Set if below
0011	SETAE/SETNB	C = 0	Set if above or equal
0100	SETE/SETZ	Z = 1	Set if equal/zero
0101	SETNE/SETNZ	Z = 0	Set if not equal/zero
0110	SETBE/SETNA	C = 1 + Z = 1	Set if below or equal
0111	SETA/SETNBE	C = 0 • Z = 0	Set if above
1000	SETS	S = 1	Set if sign
1001	SETNS	S = 0	Set if no sign
1010	SETP/SETPE	P = 1	Set if parity
1011	SETNP/SETPO	P = 0	Set if no parity
1100	SETL/SETNGE	S • O	Set if less than
1101	SETGE/SETNL	S = 0	Set if greater than or equal
1110	SETLE/SETNG	Z = 1 + S • O	Set if less than or equal
1111	SETG/SETNLE	Z = 0 + S = O	Set if greater than

SGDT/SIDT/SLDT Store descriptor table registers			
00001111 00000001 oo000mmm disp			
Format	Examples	Microprocessor	Clocks
SGDT mem	SGDT MEMORY SGDT GLOBAL	8086	—
		8088	—
		80286	11
		80386	9
		80486	10
		Pentium–Core2	4
00001111 00000001 oo001mmm disp			
Format	Examples	Microprocessor	Clocks
SIDT mem	SIDT DATAS SIDT INTERRUPT	8086	—
		8088	—
		80286	12
		80386	9
		80486	10
		Pentium–Core2	4
00001111 00000000 oo000mmm disp			
Format	Examples	Microprocessor	Clocks
SLDT reg	SLDT CX SLDT DX	8086	—
		8088	—
		80286	2
		80386	2
		80486	2
		Pentium–Core2	2
SLDT mem	SLDT NUMBS SLDT LOCALS	8086	—
		8088	—
		80286	3
		80386	2
		80486	3
		Pentium–Core2	2

SHLD/SHRD Double precision shift

00001111 10100100 oorrrmmm disp data		O D I T S Z A P C ? * * ? * *	
Format	Examples	Microprocessor	Clocks
SHLD reg,reg,imm	SHLD AX,CX,10 SHLD DX,BX,8 SHLD CX,DX,2	8086	—
		8088	—
		80286	—
		80386	3
		80486	2
		Pentium–Core2	4
SHLD mem,reg,imm	SHLD DATAQ,CX,8	8086	—
		8088	—
		80286	—
		80386	7
		80486	3
		Pentium–Core2	4
00001111 10101100 oorrrmmm disp data			
Format	Examples	Microprocessor	Clocks
SHRD reg,reg,imm	SHRD CX,DX,2	8086	—
		8088	—
		80286	—
		80386	3
		80486	2
		Pentium–Core2	4
SHRD mem,reg,imm	SHRD DATAZ,DX,4	8086	—
		8088	—
		80286	—
		80386	7
		80486	2
		Pentium–Core2	4

00001111 10100101 oorrrmmm disp Format	Examples	Microprocessor	Clocks
SHLD reg,reg,CL	SHLD BX,DX,CL	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	4 or 5
SHLD mem,reg,CL	SHLD DATAZ,DX,CL	8086	—
		8088	—
		80286	—
		80386	7
		80486	3
		Pentium–Core2	4 or 5
00001111 10101101 oorrrmmm disp Format	Examples	Microprocessor	Clocks
SHRD reg,reg,CL	SHRD AX,DX,CL	8086	—
		8088	—
		80286	—
		80386	3
		80486	3
		Pentium–Core2	4 or 5
SHRD mem,reg,CL	SHRD DATAZ,DX,CL	8086	—
		8088	—
		80286	—
		80386	7
		80486	3
		Pentium–Core2	4 or 5

SMSW Store machine status word (80286)

00001111 00000001 oo100mmm disp

Format	Examples	Microprocessor	Clocks
SMSW reg	SMSW AX SMSW DX SMSW BP	8086	—
		8088	—
		80286	2
		80386	10
		80486	2
		Pentium–Core2	4
SMSW mem	SMSW DATAQ	8086	—
		8088	—
		80286	3
		80386	3
		80486	3
		Pentium–Core2	4

STC Set carry flag

11111001

O D I T S Z A P C
1

Example	Microprocessor	Clocks
STC	8086	2
	8088	2
	80286	2
	80386	2
	80486	2
	Pentium–Core2	2

STD Set direction flag

11111101

O D I T S Z A P C
1

Example	Microprocessor	Clocks
STD	8086	2
	8088	2
	80286	2
	80386	2
	80486	2
	Pentium–Core2	2

STI Set interrupt flag

11111011

O D I T S Z A P C
1

Example	Microprocessor	Clocks
STI	8086	2
	8088	2
	80286	2
	80386	3
	80486	5
	Pentium–Core2	7

STOS Store string data

1010101w

Format	Examples	Microprocessor	Clocks
STOSB STOSW STOSD	STOSB STOSW STOSD STOS DATA_LIST REP STOSB	8086	11
		8088	15
		80286	3
		80386	40
		80486	5
		Pentium–Core2	3

STR Store task register

00001111 00000000 oo001mmm disp

Format	Examples	Microprocessor	Clocks
STR reg	STR AX STR DX STR BP	8086	—
		8088	—
		80286	2
		80386	2
		80486	2
		Pentium–Core2	2
STR mem	STR DATA3	8086	—
		8088	—
		80286	2
		80386	2
		80486	2
		Pentium–Core2	2

SUB Subtract

000101dw oorrrmmm disp

O	D	I	T	S	Z	A	P	C
*				*	*	*	*	*

Format	Examples	Microprocessor	Clocks
SUB reg,reg	SUB CL,DL SUB AX,DX SUB CH,CL SUB EAX,EBX SUB ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2

SUB mem,reg	SUB DATAJ,CL SUB BYTES,CX SUB NUMBS,ECX SUB [EAX],CX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3
SUB reg,mem	SUB CL,DATAL SUB CX,BYTES SUB ECX,NUMBS SUB DX,[EBX+EDI]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	7
		80486	2
		Pentium–Core2	1 or 2
100000sw oo101mmm disp data Format	Examples	Microprocessor	Clocks
SUB reg,imm	SUB CX,3 SUB DI,1AH SUB DL,34H SUB EDX,1345H SUB CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3
SUB mem,imm	SUB DATAS,3 SUB BYTE PTR[EDI],1AH SUB DADDY,34H SUB LIST,'A' SUB TOAD,1834H	8086	17 + ea
		8088	25 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3

0010110w data Format	Examples	Microprocessor	Clocks
SUB acc,imm	SUB AL,3 SUB AX,1AH SUB EAX,34H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

TEST Test operands (logical compare)

1000001w oorrrmmm disp Format	Examples	O D I T S Z A P C 0 * * ? * 0 Microprocessor	Clocks
TEST reg,reg	TEST CL,DL TEST BX,DX TEST DH,CL TEST EBP,EBX TEST EAX,EDI	8086	5
		8088	5
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2
TEST mem,reg reg,mem	TEST DATAJ,CL TEST BYTES,CX TEST NUMBS,ECX TEST [EAX],CX TEST CL,POPS	8086	9 + ea
		8088	13 + ea
		80286	6
		80386	5
		80486	2
		Pentium–Core2	1 or 2

1111011sw oo000mmm disp data Format	Examples	Microprocessor	Clocks
TEST reg,imm	TEST BX,3 TEST DI,1AH TEST DH,44H TEST EDX,1AB345H TEST SI,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 2
TEST mem,imm	TEST DATAS,3 TEST BYTE PTR[EDI],1AH TEST DADDY,34H TEST LIST,'A' TEST TOAD,1834H	8086	11 + ea
		8088	11 + ea
		80286	6
		80386	5
		80486	2
		Pentium–Core2	1 or 2
1010100w data Format	Examples	Microprocessor	Clocks
TEST acc,imm	TEST AL,3 TEST AX,1AH TEST EAX,34H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

VERR/VERW Verify read/write

00001111 00000000 oo100mmm disp — O D I T S Z A P C (* Z)

Format	Examples	Microprocessor	Clocks
VERR reg	VERR CX VERR DX VERR DI	8086	—
		8088	—
		80286	14
		80386	10
		80486	11
		Pentium–Core2	7
VERR mem	VERR DATAJ VERR TESTB	8086	—
		8088	—
		80286	16
		80386	11
		80486	11
		Pentium–Core2	7

00001111 00000000 oo101mmm disp

Format	Examples	Microprocessor	Clocks
VERW reg	VERW CX VERW DX VERW DI	8086	—
		8088	—
		80286	14
		80386	15
		80486	11
		Pentium–Core2	7
VERW mem	VERW DATAJ VERW TESTB	8086	—
		8088	—
		80286	16
		80386	16
		80486	11
		Pentium–Core2	7

WAIT Wait for coprocessor		
10011011 Example	Microprocessor	Clocks
WAIT FWAIT	8086	4
	8088	4
	80286	3
	80386	6
	80486	6
	Pentium–Core2	1

WBINVD Write-back cache invalidate data cache		
00001111 00001001 Example	Microprocessor	Clocks
WBINVD	8086	—
	8088	—
	80286	—
	80386	—
	80486	5
	Pentium–Core2	2000+

WRMSR Write to model specific register		
00001111 00110000 Example	Microprocessor	Clocks
WRMSR	8086	—
	8088	—
	80286	—
	80386	—
	80486	—
	Pentium–Core2	30–45

XADD Exchange and add

00001111 1100000w 11rrrrrr

O D I T S Z A P C
* * * * * *

Format	Examples	Microprocessor	Clocks
XADD reg,reg	XADD EBX,ECX XADD EDX,EAX XADD EDI,EBP	8086	—
		8088	—
		80286	—
		80386	—
		80486	3
		Pentium–Core2	3 or 4

00001111 1100000w oorrrmmm disp

Format	Examples	Microprocessor	Clocks
XADD mem,reg	XADD DATA5,ECX XADD [EBX],EAX XADD [ECX+4],EBP	8086	—
		8088	—
		80286	—
		80386	—
		80486	4
		Pentium–Core2	3 or 4

XCHG Exchange

1000011w oorrrmmm

Format	Examples	Microprocessor	Clocks
XCHG reg,reg	XCHG CL,DL XCHG BX,DX XCHG DH,CL XCHG EBP,EBX XCHG EAX,EDI	8086	4
		8088	4
		80286	3
		80386	3
		80486	3
		Pentium–Core2	3

XCHG mem,reg reg,mem	XCHG DATAJ,CL XCHG BYTES,CX XCHG NUMBS,ECX XCHG [EAX],CX XCHG CL,POPS	8086	17 + ea
		8088	25 + ea
		80286	5
		80386	5
		80486	5
		Pentium–Core2	3
10010reg Format	Examples	Microprocessor	Clocks
XCHG acc,reg reg,acc	XCHG BX,AX XCHG AX,DI XCHG DH,AL XCHG EDX,EAX XCHG SI,AX	8086	3
		8088	3
		80286	3
		80386	3
		80486	3
		Pentium–Core2	2

XLAT Translate

11010111 Example	Microprocessor	Clocks
XLAT	8086	11
	8088	11
	80286	5
	80386	3
	80486	4
	Pentium–Core2	4

XOR Exclusive-OR			
000110dw oorrrmmm disp		O D I T S Z A P C 0 * * ? * 0	
Format	Examples	Microprocessor	Clocks
XOR reg,reg	XOR CL,DL XOR AX,DX XOR CH,CL XOR EAX,EBX XOR ESI,EDI	8086	3
		8088	3
		80286	2
		80386	2
		80486	1
		Pentium–Core2	1 or 2
XOR mem,reg	XOR DATAJ,CL XOR BYTES,CX XOR NUMBS,ECX XOR [EAX],CX	8086	16 + ea
		8088	24 + ea
		80286	7
		80386	6
		80486	3
		Pentium–Core2	1 or 3
XOR reg,mem	XOR CL,DATAL XOR CX,BYTES XOR ECX,NUMBS XOR DX,[EBX+EDI]	8086	9 + ea
		8088	13 + ea
		80286	7
		80386	7
		80486	2
		Pentium–Core2	1 or 2
100000sw oo110mmm disp data			
Format	Examples	Microprocessor	Clocks
XOR reg,imm	XOR CX,3 XOR DI,1AH XOR DL,34H XOR EDX,1345H XOR CX,1834H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1 or 3

XOR mem,imm	XOR DATAS,3 XOR BYTE PTR[EDI],1AH XOR DADDY,34H XOR LIST,'A' XOR TOAD,1834H	8086	17 + ea
		8088	25 + ea
		80286	7
		80386	7
		80486	3
		Pentium–Core2	1 or 3
0010101w data Format	Examples	Microprocessor	Clocks
XOR acc,imm	XOR AL,3 XOR AX,1AH XOR EAX,34H	8086	4
		8088	4
		80286	3
		80386	2
		80486	1
		Pentium–Core2	1

SIMD INSTRUCTION SET SUMMARY

The SIMD (single-instruction, multiple data) instructions add a new dimension to the use of the microprocessor for performing multimedia and other operations. The XMM registers are numbered from XMM_0 to XMM_7 and are each 128 bits in width. Data formats stored in the XMM registers and used by the SIMD instructions appear in Figure B–1.

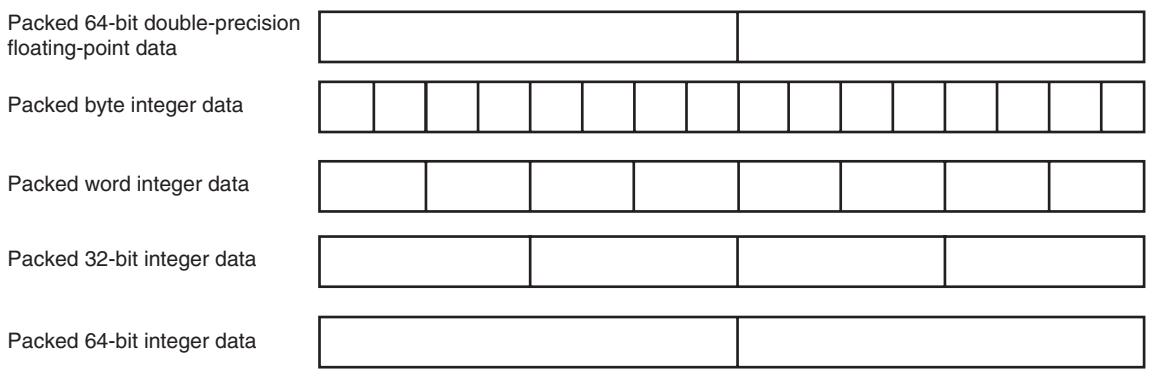


FIGURE B–1 Data formats for the 128-bit-wide XMM registers in the Pentium III and Pentium 4 microprocessors.

Data stored in the memory must be stored as 16-byte-long data in a series of memory locations accessed by using the OWORD PTR override when addressed by an instruction. The OWORD PTR override is used to address an octalword of data or 16 bytes. The SIMD instructions allow operations on packed and scalar double-precision floating-point numbers. The operation of both forms is illustrated in Figure B–2, which shows both packed and scalar multiplication. Notice that scalar only copies the leftmost double-precision number into the destination register and does not use the leftmost number in the source. The scalar instructions are meant to be compatible with the floating-point coprocessor instructions.

This section of the appendix details many of the SIMD instructions and provides examples of their usage.

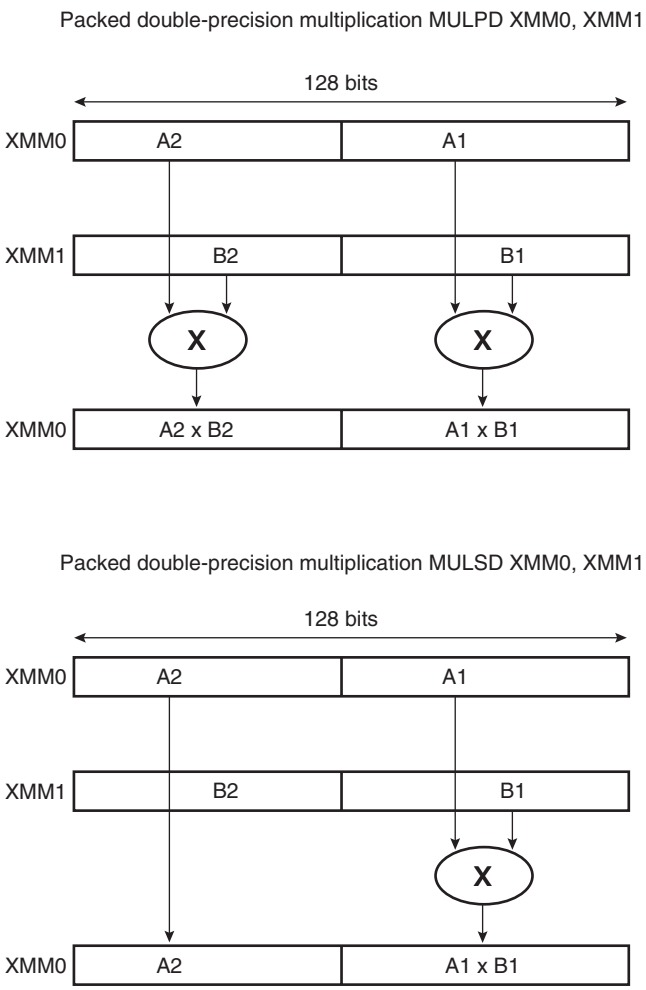


FIGURE B–2 Packed and scalar double-precision floating-point operation.

DATA MOVEMENT INSTRUCTIONS

MOVAPD	Move aligned packed double-precision data, data must be aligned on 16-byte boundaries

Examples

```
MOVAPD  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVAPD  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVUPD	Move unaligned packed double-precision data

Examples

```
MOVUPD  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVUPD  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVSD	Move scalar packed double-precision data to low quadword

Examples

```
MOVSD  XMM0, DWORD DATA3         ;copies DATA3 to XMM0
MOVSD  DWORD PTR DATA4, XMM2   ;copies XMM4 to DATA4
```

MOVHPD	Move packed double-precision data to high quadword

Examples

```
MOVHPD  XMM0, DWORD DATA3        ;copies DATA3 to XMM0
MOVHPD  DWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVLPD	Move packed double-precision data into low quadword

Examples

```
MOVLPD  XMM0, DWORD DATA3        ;copies DATA3 to XMM0
MOVLPD  DWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVMSKPD	Move packed double-precision mask

Examples

```
MOVMSKPD    EAX, XMM1    ;copies 2 sign bits to general-purpose register
```

MOVAPS	Move 4 aligned packed single-precision data, data must be aligned on 16-byte boundaries

Examples

```
MOVAPS  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVAPS  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVUPS	Move 4 unaligned packed single-precision data

Examples

```
MOVUPS  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVUPS  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVLPS	Move 2 packed single-precision numbers to low-order quadword

Examples

```
MOVLPS  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVLPS  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVHPS	Move packed single-precision numbers to high-order quadword

Examples

```
MOVHPS  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVHPS  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVAPD	Move aligned packed double-precision data, data must be aligned on 16-byte boundaries

Examples

```
MOVAPD  XMM0, OWORD DATA3        ;copies DATA3 to XMM0
MOVAPD  OWORD PTR DATA4, XMM2 ;copies XMM4 to DATA4
```

MOVLHPS	Move 2 packed single-precision numbers from the low-order quadword to the high-order quadword

Examples

```
MOVLHPS  XMM0, XMM1              ;copies XMM1 low to XMM0 high
MOVLHPS  XMM3, XMM2              ;copies XMM2 low to XMM3 high
```

MOVHLPS	Move 2 packed single-precision numbers from high-order quadword to low-order quadword

Examples

```
MOVHLPS  XMM0, XMM2          ;copies high XMM2 to low XMM0
MOVHLPS  XMM4, XMM5          ;copies high XMM5 to low XMM4
```

MOVMSKPS	Move 4-sign bits of 4 packed single-precision numbers to general-purpose register

Examples

```
MOVMSKPS  EBX, XMM0          ;copies sign bits of XMM0 to EBX
MOVMSKPS  EDX, XMM2          ;copies sign bits of XMM2 to EDX
```

ARITHMETIC INSTRUCTIONS

ADDPD	Adds packed double-precision data

Examples

```
ADDPD    XMM0, OWORD DATA3   ;adds DATA3 to XMM0
ADDPD    XMM2, XMM3          ;adds XMM3 to XMM2
```

ADDSD	Adds scalar double-precision data

Examples

```
ADDSD    XMM0, OWORD DATA3   ;adds DATA3 to XMM0
ADDSD    XMM4, XMM2          ;adds XMM2 to XMM4
```

ADDPS	Adds 2 packed single-precision numbers

Examples

```
ADDPS    XMM0, QWORD DATA3   ;adds DATA3 to XMM0
ADDPS    XMM3, XMM2          ;adds XMM2 to XMM3
```

ADDLS Adds scalar single-precision data

Examples

```
ADDLS   XMM0, DWORD DATA3      ;adds DATA3 to XMM0
ADDLS   XMM7, XMM2             ;adds XMM2 to XMM7
```

SUBPD Subtracts packed double-precision data

Examples

```
SUBPD   XMM0, OWORD DATA3      ;subtracts DATA3 from XMM0
SUBPD   XMM2, XMM3             ;subtracts XMM3 from XMM2
```

SUBSD Subtracts scalar double-precision data

Examples

```
SUBSD   XMM0, OWORD DATA3      ;subtracts DATA3 from XMM0
SUBSD   XMM4, XMM2             ;subtracts XMM2 from XMM4
```

SUBPS Subtracts 2 packed single-precision numbers

Examples

```
SUBPS   XMM0, QWORD DATA3      ;subtracts DATA3 from XMM0
SUBPS   XMM3, XMM2             ;subtracts XMM2 from XMM3
```

SUBLS Subtracts scalar single-precision data

Examples

```
SUBLS   XMM0, DWORD DATA3      ;subtracts DATA3 from XMM0
SUBLS   XMM7, XMM2             ;subtracts XMM2 from XMM7
```

MULPD Multiplies packed double-precision data

Examples

```
MULPD   XMM0, OWORD DATA3      ;multiplies DATA3 times XMM0
MULPD   XMM3, XMM2             ;multiplies XMM2 times XMM3
```

MULSD	Multiplies scalar double-precision data

Examples

```
MULSD    XMM0, OWORD DATA3      ;multiplies DATA3 times XMM0
MULSD    XMM3, XMM6             ;multiplies XMM6 times XMM3
```

MULPS	Multiplies 2 packed single-precision numbers

Examples

```
MULPS    XMM0, QWORD DATA3      ;multiplies DATA3 times XMM0
MULPS    XMM0, XMM2             ;multiplies XMM2 times XMM0
```

MULSS	Multiplies a single-precision number

Examples

```
MULSS    XMM0, DWORD DATA3      ;multiplies DATA3 times XMM0
MULSS    XMM1, XMM2             ;multiplies XMM2 times XMM1
```

DIVPD	Divides packed double-precision data

Examples

```
DIVPD    XMM0, OWORD DATA3      ;divides XMM0 by DATA3
DIVPD    XMM3, XMM2             ;divides XMM3 by XMM2
```

DIVSD	Divides scalar double-precision data

Examples

```
DIVSD    XMM0, OWORD DATA3      ;divides XMM0 by DATA3
DIVSD    XMM3, XMM6             ;divides XMM3 by XMM6
```

DIVPS	Divides 2 packed single-precision numbers

Examples

```
DIVPS    XMM0, QWORD DATA3      ;divides XMM0 by DATA3
DIVPS    XMM0, XMM2             ;divides XMM0 by XMM2
```

DIVSS Divides a single-precision number

Examples

```
DIVSS   XMM0, DWORD DATA3   ;divides XMM0 by DATA3
DIVSS   XMM1, XMM2          ;divides XMM1 by XMM2
```

SQRTPD Finds the square root of packed double-precision data

Examples

```
SQRTPD  XMM0, OWORD DATA3   ;finds square root of DATA3, result to XMM0
SQRTPD  XMM3, XMM2          ;finds square root of XMM2, result to XMM3
```

SQRTSD Finds the square root of scalar double-precision data

Examples

```
SQRTSD  XMM0, OWORD DATA3   ;finds square root of DATA3, result to XMM0
SQRTSD  XMM3, XMM6          ;finds square root of XMM6, result to XMM3
```

SQRTPS Finds the square root of 2 packed single-precision numbers

Examples

```
SQRTPS  XMM0, QWORD DATA3   ;finds square root of DATA3, result to XMM0
SQRTPS  XMM0, XMM2          ;finds square root of XMM2, result to XMM0
```

SQRTSS Finds the square root of a single-precision number

Examples

```
SQRTSS  XMM0, DWORD DATA3   ;finds the square root of DATA3, result to XMM0
SQRTSS  XMM1, XMM2          ;finds the square root of XMM2, result to XMM1
```

RCPPS Finds the reciprocal of a packed single-precision number

Examples

```
RCPPS   XMM0, OWORD DATA3   ;finds the reciprocal of DATA3, result to XMM0
RCPPS   XMM3, XMM2          ;finds the reciprocal of XMM2, result to XMM3
```

RCPSS	Finds the reciprocal of a single-precision number

Examples

```
RCPSS    XMM0, OWORD DATA3    ;finds the reciprocal of DATA3, result to XMM0
RCPSS    XMM3, XMM6           ;finds the reciprocal of XMM6, result to XMM3
```

RSQRTPS	Finds reciprocals of packed single-precision data

Examples

```
RSQRTPS  XMM0, OWORD DATA3    ;finds reciprocal of square root of DATA3
RSQRTPS  XMM3, XMM2           ;finds reciprocal of square root of XMM2
```

RSQRTSS	Finds the reciprocal of square root of a scalar single-precision number

Examples

```
RSQRTSS  XMM0, OWORD DATA3    ;finds reciprocal of square root of DATA3
RSQRTSS  XMM3, XMM6           ;finds reciprocal of square root of XMM6
```

MAXPD	Compares and returns the maximum packed double-precision floating-point number

Examples

```
MAXPD    XMM0, OWORD DATA3    ;compares numbers in DATA3, largest to XMM0
MAXPD    XMM3, XMM2           ;compares numbers in XMM2, largest to XMM3
```

MAXSD	Compares scalar double-precision data and returns the largest

Examples

```
MAXSD    XMM0, OWORD DATA3    ;compares numbers in DATA3, largest to XMM0
MAXSD    XMM3, XMM6           ;compares numbers in XMM6, largest to XMM3
```

MAXPS	Compares and returns the largest packed single-precision number

Examples

```
MAXPS    XMM0, QWORD DATA3    ;compares numbers in DATA3, largest to XMM0
MAXPS    XMM0, XMM2           ;compares numbers in XMM2, largest to XMM0
```

MAXSS	Compares scalar single-precision numbers and returns the largest

Examples

```
MAXSS   XMM0, DWORD DATA3   ;compares numbers in DATA3, largest to XMM0
MAXSS   XMM1, XMM2          ;compares numbers in XMM2, largest to XMM1
```

MINPD	Compares and returns the minimum packed double-precision floating-point number

Examples

```
MINPD   XMM0, OWORD DATA3   ;compares numbers in DATA3, least to XMM0
MINPD   XMM3, XMM2          ;compares numbers in XMM2, least to XMM3
```

MINSD	Compares scalar double-precision data and returns the smallest

Examples

```
MINSD   XMM0, OWORD DATA3   ;compares numbers in DATA3, least to XMM0
MINSD   XMM3, XMM6          ;compares numbers in XMM6, least to XMM3
```

MINPS	Compares and returns the smallest packed single-precision number

Examples

```
MINPS   XMM0, QWORD DATA3   ;compares numbers in DATA3, least to XMM0
MINPS   XMM0, XMM2          ;compares numbers in XMM2, least to XMM0
```

MINSS	Compares scalar single-precision numbers and returns the smallest

Examples

```
MINSS   XMM0, DWORD DATA3   ;compares numbers in DATA3, least to XMM0
MINSS   XMM1, XMM2          ;compares numbers in XMM2, least to XMM1
```

LOGIC INSTRUCTIONS

ANDPD ANDs packed double-precision data

Examples

```
ANDPD    XMM0, OWORD DATA3    ;ands DATA3 to XMM0
ANDPD    XMM2, XMM3           ;ands XMM3 to XMM2
```

ANDNPD NANDs packed double-precision data

Examples

```
ANDNPD   XMM0, OWORD DATA3    ;Nands DATA3 to XMM0
ANDNPD   XMM4, XMM2           ;Nands XMM2 to XMM4
```

ANDPS ANDs 2 packed single-precision data

Examples

```
ANDPS    XMM0, QWORD DATA3    ;ands DATA3 to XMM0
ANDPS    XMM3, XMM2           ;ands XMM2 to XMM3
```

ANDNPS NANDs 2 packed single-precision data

Examples

```
ANDNPS   XMM0, DWORD DATA3    ;Nands DATA3 to XMM0
ANDNPS   XMM7, XMM2           ;Nands XMM2 to XMM7
```

ORPD ORs packed double-precision data

Examples

```
ORPD     XMM0, OWORD DATA3    ;ors DATA3 to XMM0
ORPD     XMM2, XMM3           ;ors XMM3 to XMM2
```

ORPS ORS 2 packed single-precision numbers

Examples

```
ORPS     XMM0, OWORD DATA3    ;ors DATA3 to XMM0
ORPS     XMM3, XMM2           ;ors XMM2 to XMM3
```

XORPD	Exclusive-ORs packed double-precision data	
Examples		
XORPD	XMM0, OWORD DATA3	;exclusive-ors DATA3 to XMM0
XORPD	XMM2, XMM3	;exclusive-ors XMM3 to XMM2

XORPS	Exclusive-ORs packed double-precision data	
Examples		
XORPS	XMM0, OWORD DATA3	;exclusive-ors DATA3 to XMM0
XORPS	XMM2, XMM3	;exclusive-ors XMM3 to XMM2

COMPARISION INSTRUCTIONS

CMPPD	Compares packed double-precision numbers	
Examples		
CMPPD	XMM0, OWORD DATA3	;compares DATA3 with XMM0
CMPPD	XMM2, XMM3	;compares XMM3 with XMM2

CMPSD	Compares scalar double-precision data	
Examples		
CMPSD	XMM0, QWORD DATA3	;compares DATA3 with XMM0
CMPSD	XMM3, XMM2	;compares XMM2 with XMM3

CMPISD	Compares scalar double-precision data and sets EFAGS	
Examples		
CMPISD	XMM0, OWORD DATA3	;compares DATA3 with XMM0
CMPISD	XMM2, XMM3	;compares XMM3 with XMM2

UCOMISD	Compares scalar unordered double-precision numbers and changes EFLAGS

Examples

```
UCOMISD   XMM0, QWORD DATA3   ;compares DATA3 with XMM0
UCOMISD   XMM3, XMM2          ;compares XMM2 with XMM3
```

CMPPS	Compares packed single-precision data

Examples

```
CMPPS   XMM0, OWORD DATA3   ;compares DATA3 with XMM0
CMPPS   XMM2, XMM3          ;compares XMM3 with XMM2
```

CMPSS	Compares 2 packed single-precision numbers

Examples

```
CMPSS   XMM0, QWORD DATA3   ;compares DATA3 with XMM0
CMPSS   XMM3, XMM2          ;compares XMM2 with XMM3
```

COMISS	Compares scalar single-precision data and changes EFLAGS

Examples

```
COMISS   XMM0, OWORD DATA3   ;compares DATA3 with XMM0
COMISS   XMM2, XMM3          ;compares XMM3 with XMM2
```

UCOMISS	Compares unordered single-precision numbers and changes EFLAGS

Examples

```
UCOMISS   XMM0, QWORD DATA3   ;compares DATA3 with XMM0
UCOMISS   XMM3, XMM2          ;compares XMM2 with XMM3
```

DATA CONVERSION INSTRUCTIONS

SHUFPD Shuffles packed double-precision numbers

Examples

```
SHUFPD    XMM0, OWORD DATA3    ;shuffles DATA3 with XMM0
SHUFPD    XMM2, XMM2           ;swaps upper and lower quadword in XMM2
```

UNPCKHPD Unpacks the upper double-precision number

Examples

```
UNPCKHPD  XMM0, OWORD DATA3    ;unpacks DATA3 into XMM0
UNPCKHPD  XXM3, XMM2           ;unpacks XMM2 into XMM3
```

UNPCKLPD Unpacks the lower double-precision number

Examples

```
UNPCKLPD  XMM0, OWORD DATA3    ;unpacks DATA3 into XMM0
UNPCKLPD  XMM3, XMM2           ;unpacks XMM2 into XMM3
```

SHUFPS Shuffles packed single-precision numbers

Examples

```
SHUFPS    XMM0, QWORD DATA3    ;shuffles DATA3 with XMM0
SHUFPS    XMM2, XMM2           ;swaps upper and lower quadword in XMM2
```

UNPCKHPS Unpacks the lower double-precision number

Examples

```
UNPCKHPS  XMM0, QWORD DATA3    ;unpacks DATA3 into XMM0
UNPCKHPS  XMM3, XMM2           ;unpacks XMM2 into XMM3
```

UNPCKLPSD Unpacks the lower double-precision number

Examples

```
UNPCKLPSD  XMM0, QWORD DATA3   ;unpacks DATA3 into XMM0
UNPCKLPSD  XXM3, XMM2          ;unpacks XMM2 into XMM3
```

APPENDIX C

Flag-Bit Changes

This appendix shows only the instructions that actually change the flag bits. Any instruction not listed does not affect any of the flag bits.

Instruction	*O*	*D*	*I*	*T*	*S*	*Z*	*A*	*P*	*C*
AAA	?				?	?	*	?	*
AAD	?				*	*	?	*	?
AAM	?				*	*	?	*	?
AAS	?				?	?	*	?	*
ADC	*				*	*	*	*	*
ADD	*				*	*	*	*	*
AND	0				*	*	?	*	0
ARPL						*			
BSF						*			
BSR						*			
BT									*
BTC									*
BTR									*
BTS									*
CLC									0
CLD		0							
CLI			0						
CMC									*
CMP	*				*	*	*	*	*
CMPS	*				*	*	*	*	*
CMPXCHG	*				*	*	*	*	*
CMPXCHG8B						*			
DAA	?				*	*	*	*	*
DAS	?				*	*	*	*	*
DEC	*				*	*	*	*	
DIV	?				?	?	?	?	?
IDIV	?				?	?	?	?	?
IMUL	*				?	?	?	?	*
INC	*				*	*	*	*	

Instruction	*O*	*D*	*I*	*T*	*S*	*Z*	*A*	*P*	*C*
IRET	*	*	*	*	*	*	*	*	*
LAR						*			
LSL						*			
MUL	*				?	?	?	?	*
NEG	*				*	*	*	*	*
OR	0				*	*	?	*	0
POPF	*	*	*	*	*	*	*	*	*
RCL/RCR	*								*
REPE/REPNE						*			
ROL/ROR	*								*
SAHF					*	*	*	*	*
SAL/SAR	*				*	*	?	*	*
SHL/SHR	*				*	*	?	*	*
SBB	*				*	*	*	*	*
SCAS	*				*	*	*	*	*
SHLD/SHRD	?				*	*	?	*	*
STC									1
STD		1							
STI			1						
SUB	*				*	*	*	*	*
TEST	0				*	*	?	*	0
VERR/VERW						*			
XADD	*				*	*	*	*	*
XOR	0				*	*	?	*	0